Toyota PCS / VCH Data Outputs



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Brad Muir

- 30 Years Police Experience
  - 21 Years as Reconstructionist with Ontario Provincial Police
  - Crash Recon Unit since 1995
  - Supervised of one of 4 crash teams in the Greater Toronto Region
- Crash testing Experience
  - 400+ CDR/EDR Crash Tests
  - Conducting training in Germany, Australia, UK and all over North America





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What Data Can I get?

- Vehicle Control History
  - With Pre & Post Trigger data it may help you better understand pre- and post- collision vehicle inputs and behavior
- MAY contain images
  - Same pre/post trigger timing as VCH
  - Low-Res black & white images
- TSS => 2.0 data easily reconciled with corresponding CDR data – should have 17EDR/19EDR
- VCH may exist when event delta-v < 8km/h threshold

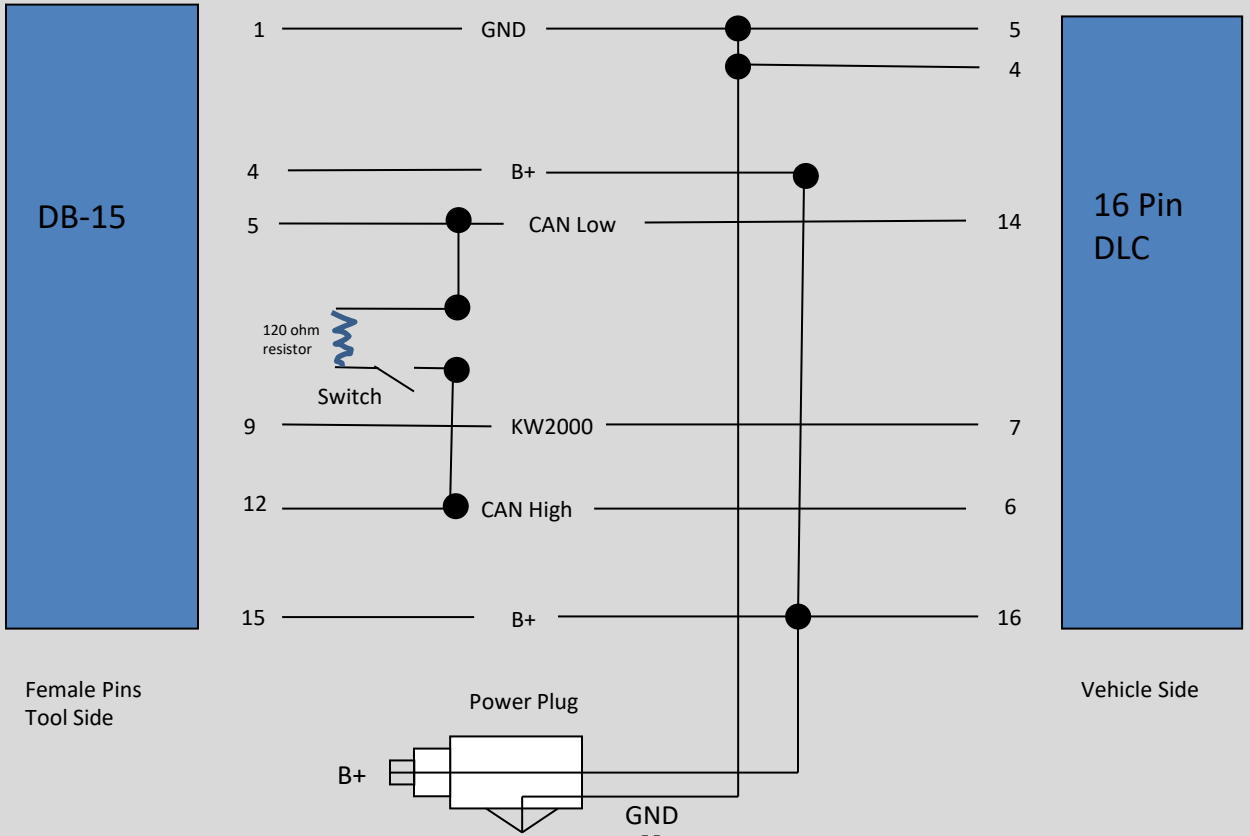
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When Should I Go After VCH Data?

- On ANY Toyota / Lexus (Scion) vehicle that supports this data!!
- Remember VCH may exist when event delta-v < 8km/h threshold
- Pedestrian Collisions
  - Usually low delta-v (for the vehicle!)
  - Driver may have braked hard (or may have accelerated hard to get away!!
- Remember VCH has PRE & POST Impact data!

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DLC to CDR Interface Cable Adapter with External Power Input – Version 1 (All CAN ACM's – not CAN FD)



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Update on VCH D2M Retrieval

- Current Cable works thru 2021 vehicles with TSS 1.0
  - Both TSS-C and TSS-P
- Vehicles with TSS =>2.0 may be CAN-FD ACM Equipped
- These will require a Network Gateway inserted between the VCI and the ACM
- Currently testing solutions with TMC and TMNA

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Update on Camera Images D2M

- FFD Data may be retrieved D2M but NOT images
- Toyota Japan supplied wiring info was not successful
- Continue to explore required components for TSS 1.0 as well as 2.0 and newer
- Currently best option is surrogate vehicle if back-power is not possible
  - Note: It appears a valid ignition key / fob signal is required
- If Surrogate – MUST be same Model Year and recommended build date be as close as possible
  - There may be different software versions in Camera and other components that will result in failure

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Be Aware: TSS 1.0

- For Images to be recorded and linked the VEHICLE must brake
- A PCS warning buzzer is not sufficient for images to be linked with VCH and recorded
- If vehicle begins to brake and driver responds, then related images should be stored
- CHECK: perform CDR download FIRST. Look for deceleration with no driver input: i.e. no brake switch = on, no brake oil pressure, but speed drops more than idling down

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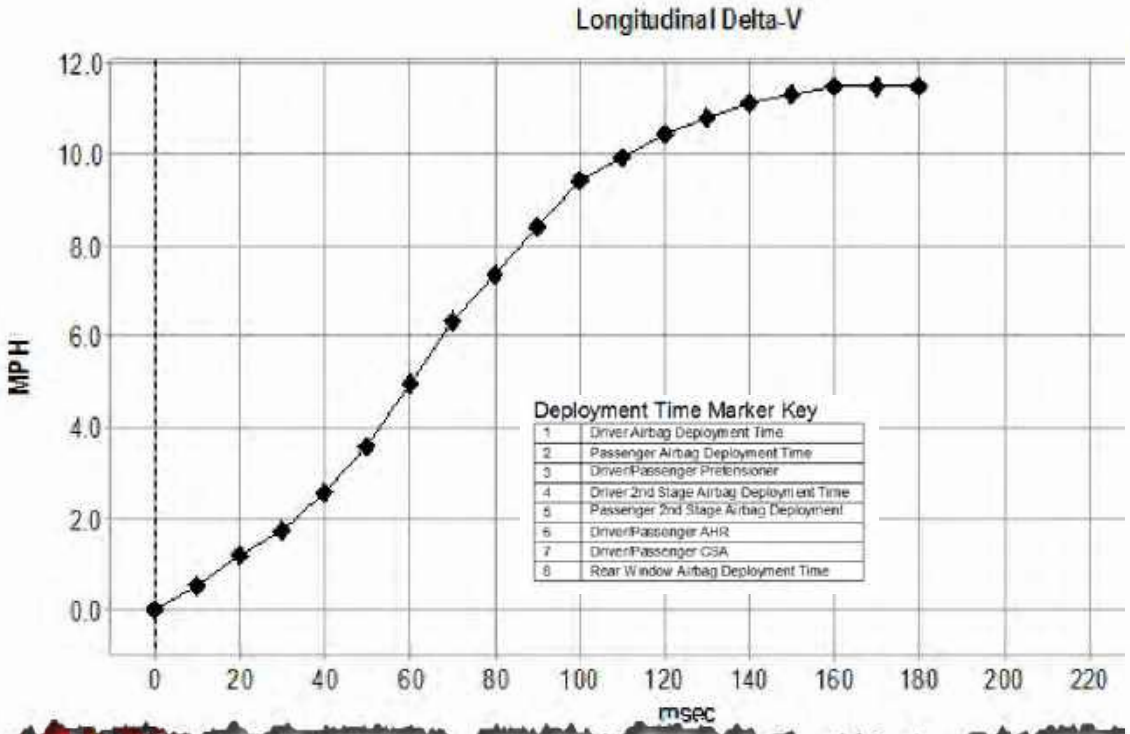
Pre-Collision Braking System





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Delta-V



Longitudinal Crash Pulse (Most Recent Event, TRG

Time (msec)	Longitudinal Delta-V (MPH [km/h])	Power Supply Status
0	0.0 [0.0]	ON
10	0.5 [0.8]	ON
20	1.2 [1.9]	ON
30	1.7 [2.8]	ON
40	2.6 [4.1]	ON
50	3.6 [5.8]	ON
60	5.0 [8.0]	ON
70	6.3 [10.2]	ON
80	7.4 [11.9]	ON
90	8.4 [13.5]	ON
100	9.4 [15.2]	ON
110	9.9 [16.0]	ON
120	10.5 [16.8]	ON
130	10.8 [17.4]	ON
140	11.1 [17.9]	ON
150	11.3 [18.2]	ON
160	11.5 [18.5]	ON
170	11.5 [18.5]	ON
180	11.5 [18.5]	ON
190	-0.2 [-0.3]	ON
200	-0.2 [-0.3]	ON
210	-0.2 [-0.3]	ON
220	-0.2 [-0.3]	ON
230	-0.2 [-0.3]	ON
240	-0.2 [-0.3]	ON
250	-0.2 [-0.3]	ON

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Pre-Crash Data

Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 1)

Time (sec)	-4.55	-4.05	-3.55	-3.05	-2.55	-2.05	-1.55	-1.05	-0.55	-0.05	0 (TRG)
Vehicle Speed (MPH [km/h])	19.9 [32]	19.9 [32]	19.9 [32]	20.5 [33]	20.5 [33]	18.6 [30]	9.3 [15]	5 [8]	0.6 [1]	0 [0]	0 [0]
Accelerator Pedal, % Full (%)	13.5	18.0	23.5	21.5	14.5	23.5	67.0	0.0	0.0	0.0	0.0
Percentage of Engine Throttle (%)	3.0	4.0	4.5	5.5	3.0	4.5	0.0	3.0	0.0	0.0	0.0
Engine RPM (RPM)	1,300	1,300	1,300	1,300	1,300	1,200	1,000	1,800	900	700	700
Service Brake, ON/OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	ON	ON	ON	ON
Brake Oil Pressure (Mpa)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.44	4.37	7.87	6.34
Longitudinal Acceleration , VSC Sensor (m/sec^2)	0.359	0.287	0.287	0.287	0.502	-4.450	-8.973	-1.005	-4.307	8.973	8.973
YawRate (deg/sec)	0.00	-0.49	0.00	0.00	-0.49	-0.49	1.46	-0.49	-0.49	0.00	3.90
Steering Input (degrees)	3	3	3	3	3	3	3	6	15	24	27
Shift Position	D	D	D	D	D	D	D	D	D	D	D

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VCH Trigger List

- Long List! – Refer to www.techinfo.Toyota.com !!

TRIGGER ITEM		RECORD DATA				
		BEFORE DETECTION		AFTER DETECTION		SAMPLING CYCLE (SEC.)
ID NO.	ITEM	RECORDING TIME (SEC.)	NUMBER OF POINTS	RECORDING TIME (SEC.)	NUMBER OF POINTS	
20-3	ICS operation history (brake control)					
20-4	ICS operation history (stop)					
20-5	ICS operation history (regression)					
20-6	ICS operation history (temporarily not available)					
22	Lane change assist operation history	8	20	0	1	0.4
23	FHL operation history	5	25	3	16	0.2
24	SCB operation history	5	34	5	35	0.15
26-1	Stop assist history (shift to deceleration phase)					
26-2	Stop assist history (shift to stop phase)					

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Trigger Parameters

General data* *(Recorded for all Triggers)		Powertrain system	Brake control system	Driver Assistance system
·Vehicle speed ·Num. of ignition cycles ·Time from IG-ON ·Odometer ·Accelerator pedal % ·Shift position ·Brake SW ·Brake fluid pressure ·Steering angle	·Fuel injection command ·Longitudinal G sensor ·Lateral G sensor ·Yaw rate	·Engine RPM ·Gear position ·Turbo RPM ·Ready-ON (HV) ·% of Engine throttle ·Engine load factor	·VSC/ABS/TRC activation ·VSC/TRC switch ·Wheel speed sensors	·Cruise control SW ·Cruise control operation ·Pre Collision System activation ·Lane Keep Assist ·Target distance ·Target relative velocity ·Drive mode

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Vehicle Control History - Parameters

- Not all parameters are shown
- Refer to Vehicle Control History section of on-line Service Manual
- Techinfo.Toyota.com*
- *Subscription Required

ITEM NAME	ITEM EXPLANATION	DISPLAYED VALUE	REMARKS
Vehicle Speed	Displays the vehicle speed.	0 to 200 km/h	-
Accelerator Opening Ratio	Displays the accelerator position sensor opening angle.	0 to 100%	-
Engine RPM Data	Displays the engine speed.	0 to 12800 rpm	-
Cruise Control	Displays whether the cruise control is operating.	ON: Operating OFF: Not Operating / Unequipped Vehicles	Unequipped vehicles display OFF.
Shift Position Signal	Displays the shift lever position.	N / D / R / P / 5 / 4 / 3 / 2 / LO / B / SD / Sequential Mode / Not Specified	-
Brake SW	Displays the stop light switch status.	ON: Stop light illuminated OFF: Not Illuminated	-
Brake Oil Pressure	Displays the fluid pressure inside the master cylinder.	0 to 12.144 MPa	-
Yaw Rate Sensor Signal	Displays the angular velocity when turning.	-61 to 61 deg/s	Becomes a positive value when turning left.
Longitudinal G Sensor Signal	Displays the front and rear acceleration detected by the sensor.	-8.9725 to 8.9725 m/s^2	• Acceleration becomes a positive value • Displays "INVALID" if below the smallest value
Lateral G Sensor Signal	Displays the right and left acceleration detected by the sensor.	-8.9725 to 8.9725 m/s^2	Becomes a positive value when turning left.
Throttle Opening Ratio	Displays the engine throttle opening amount.	0 to 100%	Only displayed for diesel vehicles ("INVALID" is displayed for other vehicles)
Command Value of Fuel Injection Amount	Displays the fuel injection volume (ECU command value).	0 to 120 mm^3/st	Only displayed for diesel vehicles.
Steering Signal	Displays the steering angle.	-567 to 567 deg	Becomes a positive value when turning left.
HV Recognition Flag		HV: Hybrid vehicle	-

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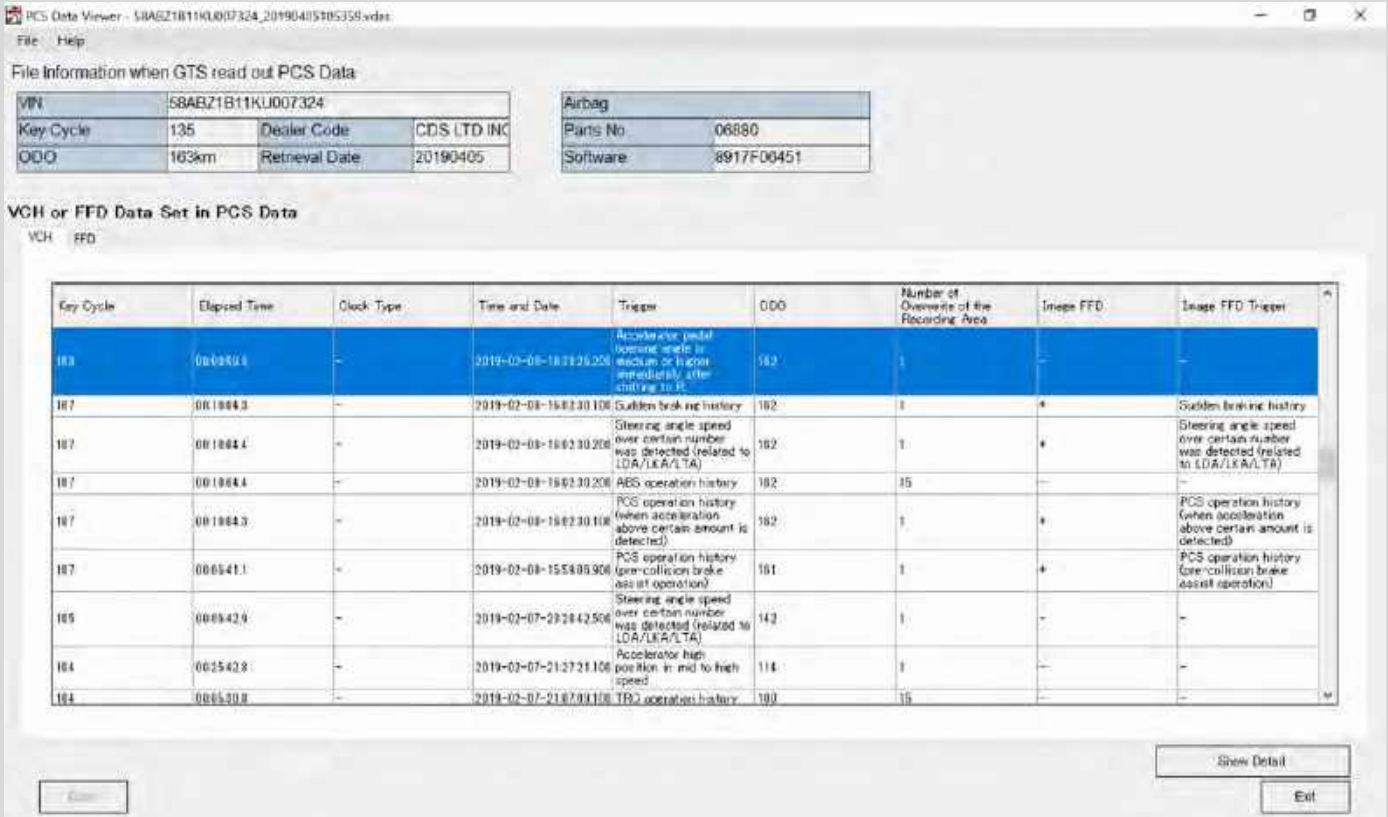
Image Recording

Image Data Specifications

TRIGGER ITEM		RECORD DATA				
		BEFORE DETECTION		AFTER DETECTION		SAMPLING CYCLE (SEC.)
ID NO.	ITEM	RECORDING TIME (SEC.)	NUMBER OF POINTS	RECORDING TIME (SEC.)	NUMBER OF POINTS	
15	Sudden braking history	6	10	6	10	0.6
16	Sudden turning history	6	10	6	10	0.6
18-3	PCS operation history (pre-collision brake assist operation)	6	10	6	10	0.6
18-4	PCS operation history (prior brake operation)					
18-5	PCS operation history (pre-collision brake operation)					
18-11	PCS operation history (when acceleration above certain amount is detected)					
18-14	Pedestrian protection system over constant pressure sensor signal input					
19	Torque sensor signal above certain amount is detected during LKA/LDA operation	6	10	6	10	0.6
20-2	ICS operation history (max output limit)	6	10	6	10	0.6
20-3	ICS operation history (brake control)					

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PCS Data Viewer



TSS => 2.0 – Event Date / Time





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TSS => 2.0 – Event Date / Time

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO
108	00:00:09.8	–	2019-02-08-16:38:26.200	Accelerator pedal opening angle is medium or higher immediately after shifting to R	162
107	00:10:04.3	–	2019-02-08-16:02:30.100	Sudden braking history	162
107	00:10:04.4	–	2019-02-08-16:02:30.200	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	162
107	00:10:04.4	–	2019-02-08-16:02:30.200	ABS operation history	162
107	00:10:04.3	–	2019-02-08-16:02:30.100	PCS operation history (when acceleration above certain amount is detected)	162
107	00:06:41.1	–	2019-02-08-15:59:06.900	PCS operation history (pre-collision brake assist operation)	161
105	00:06:42.9	–	2019-02-07-23:28:42.500	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	142
104	00:25:42.8	–	2019-02-07-21:27:21.100	Accelerator high position in mid to high speed	114
104	00:05:30.8	–	2019-02-07-21:07:09.100	TRC operation history	100

Back



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Toyota TechStream System

- Dealer Tool
  - PC Based – Toughbook with VIM
- Aftermarket Tool
  - VIM from Drew Technologies
  - Mongoose Pro MFC VIM or
  - Mongoose Pro MFC2 VIM
  - CDR900 + others
- Software Subscription from techinfo.toyota.com







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www.techinfo.Toyota.com

- Subscription basis

Subscription Type	Standard	Professional Diagnostic	Security Professional*
2 Day	**$20**	**$65**	**$65**
Monthly	**$90**	**NA**	**NA**
Yearly	**$480**	**$1295**	**$1295**

NOTE: All subscriptions expire at midnight, Pacific Time on the last day of your term.
This means a two-day subscription may actually provide greater than 48 hours of usable access.

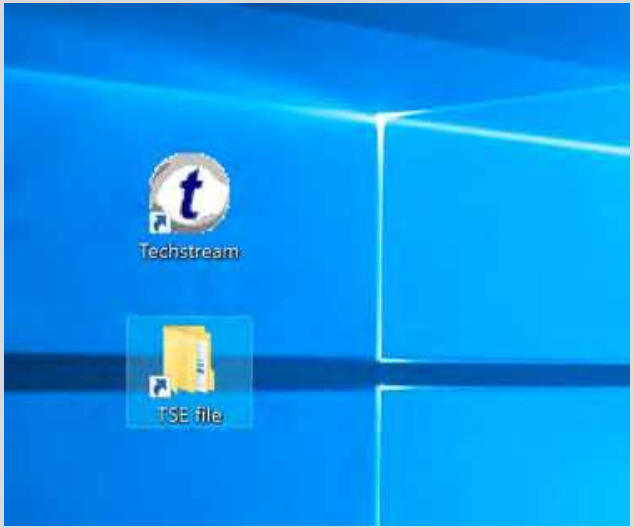
- **** PCS Data Viewer ONLY works during Subscription period!! ****

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Toyota TechStream Setup

After successful installation two new shortcuts will appear on your desktop.

You can start the application from this shortcut



A new “TechStream Diagnostics” folder appears in your Start menu

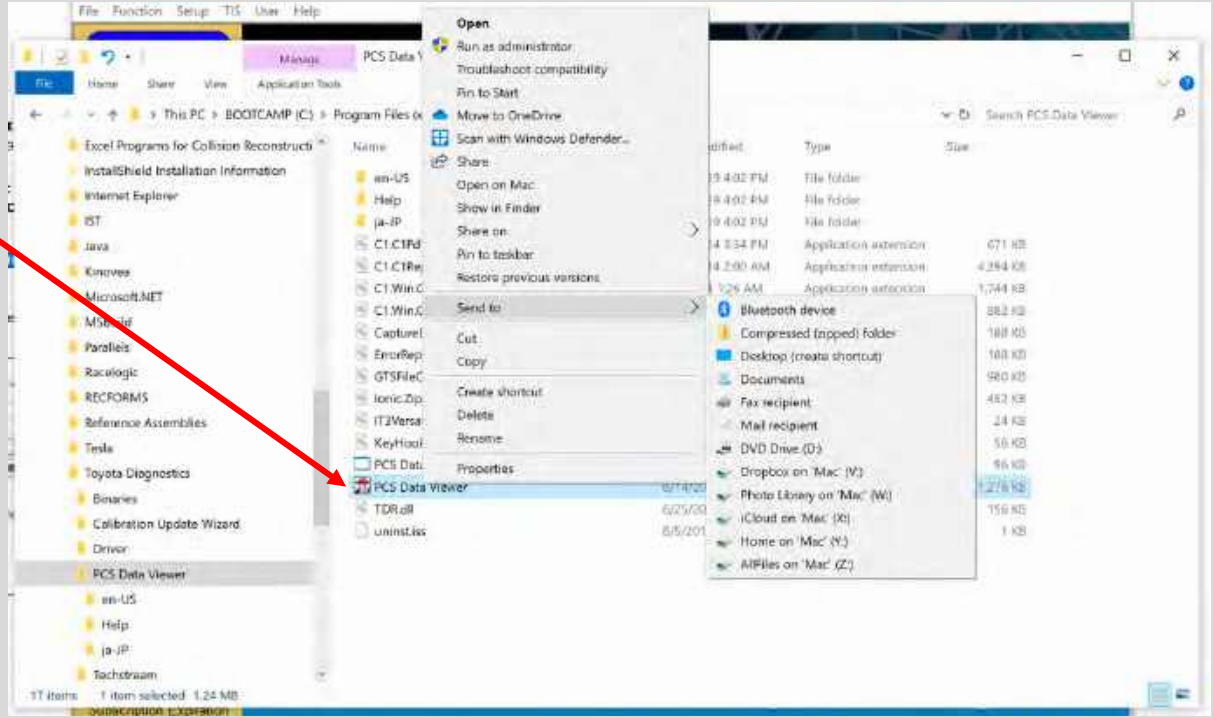
Up can launch the application from this menu



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Toyota TechStream Setup

- To access VDAS files and Camera Images retrieved in the sections that follow, you will need to launch the “PCS Data Viewer” application
- This application is installed with TechStream, and once TechStream is activated, it can be launched
- Navigate from a Windows Explorer window to here:
- Right-click “PCS Data Viewer” and click “Send To” and chose “Desktop – Create Shortcut”



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Toyota TechStream Setup

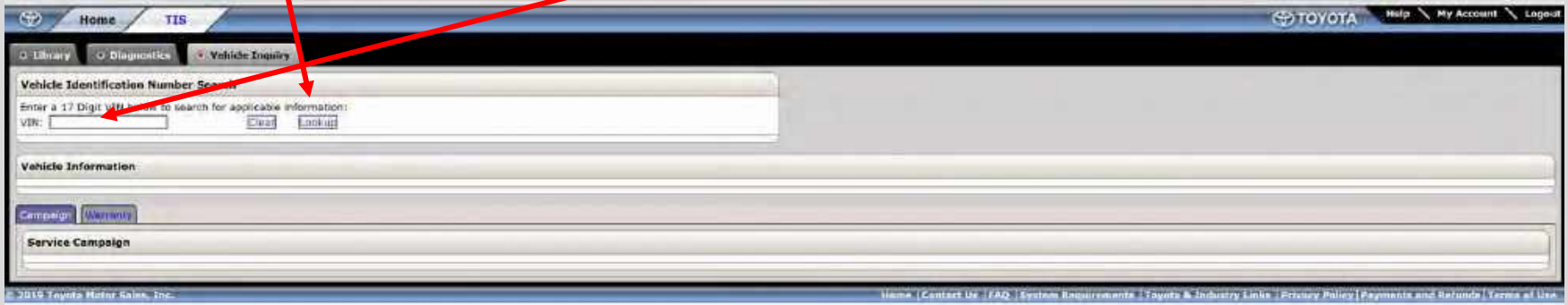
- Check your Windows Desktop
- You should see these 3 Icons:
  - PCS Data Viewer – launches the tool to view the VDAS file and create output that includes Vehicle Control History data and any related images (Depends on TSS 1.0 or 2.0 – version varies depending on make/model/year of vehicle
  - TechStream
  - TSE File – directory for information stored using TechStream



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Toyota TechStream – Vehicle Inquiry

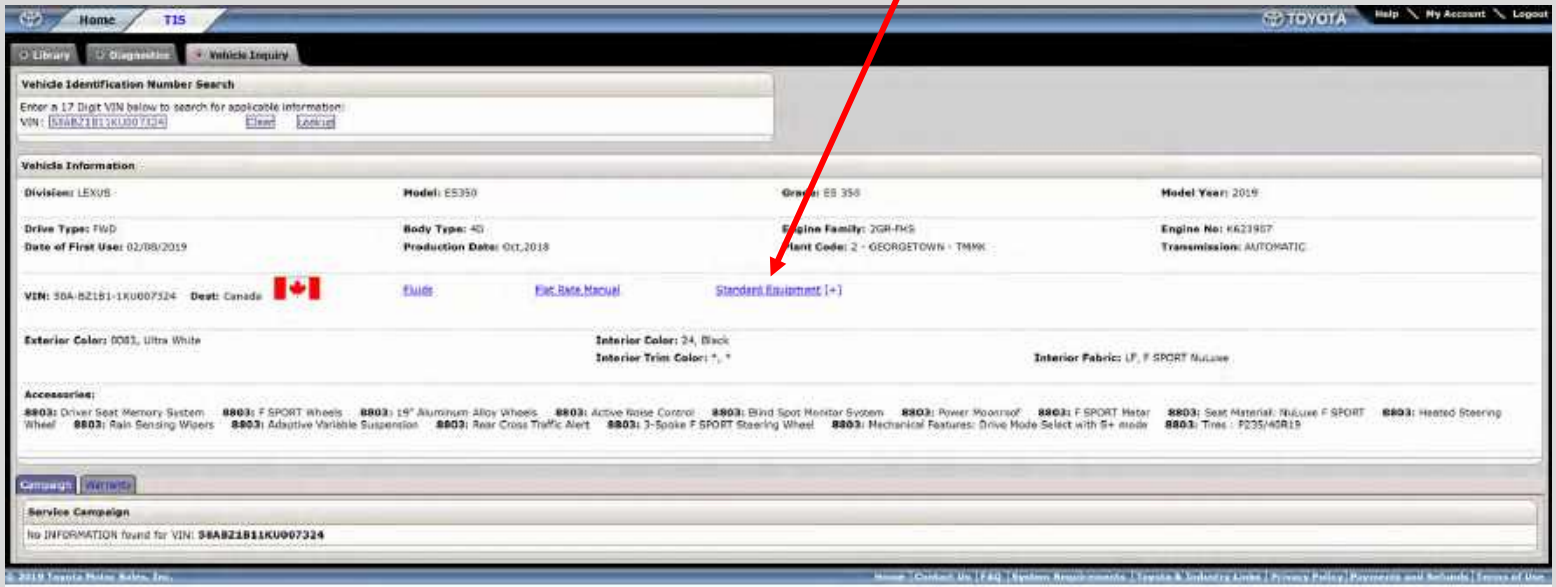
- Your subscription includes access to Service manuals, Owners Manuals, and Vehicle “as-built” information
- Click “Vehicle Inquiry” Tab and input VIN of subject Vehicle
- Click “Lookup”



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Toyota TechStream – Vehicle Inquiry

- “As-Built” record is shown. Click the “+” beside “Standard Equipment” to expand the displayed data



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Toyota TechStream – Vehicle Inquiry

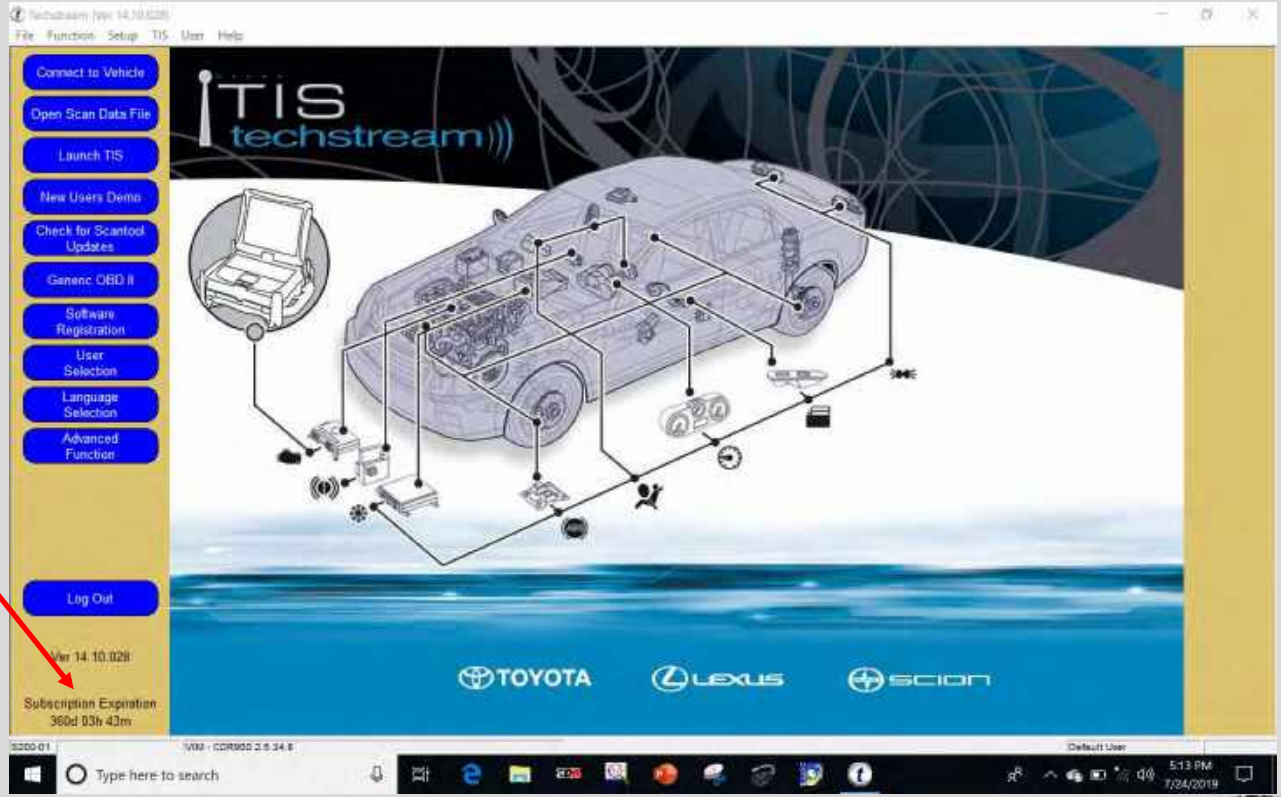
- A listing of the vehicle as-built equipment will be shown, including any optional equipment installed at the factory.
- This info can also be printed.



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Toyota TechStream Operation

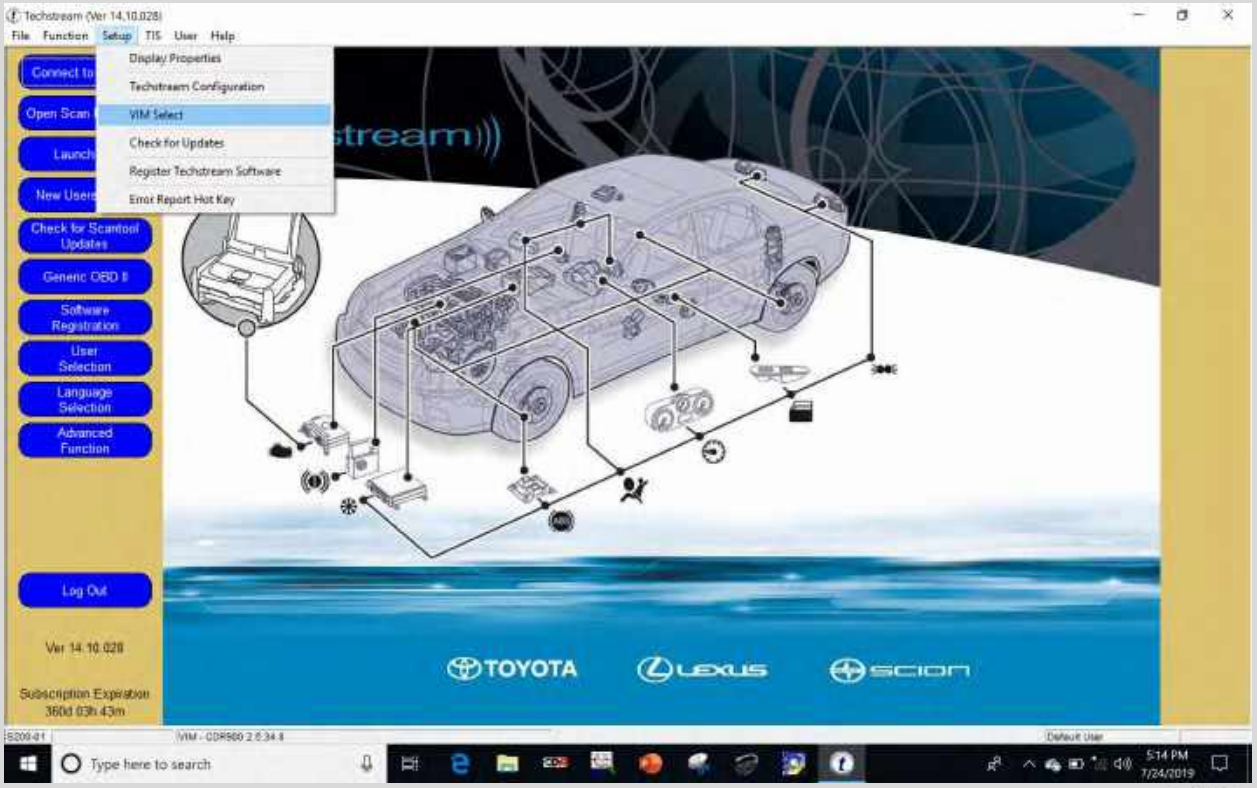
- Launch TechStream from the desktop icon
- Subscription Expiration
- Check VIM Setup



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Toyota TechStream VIM Setup

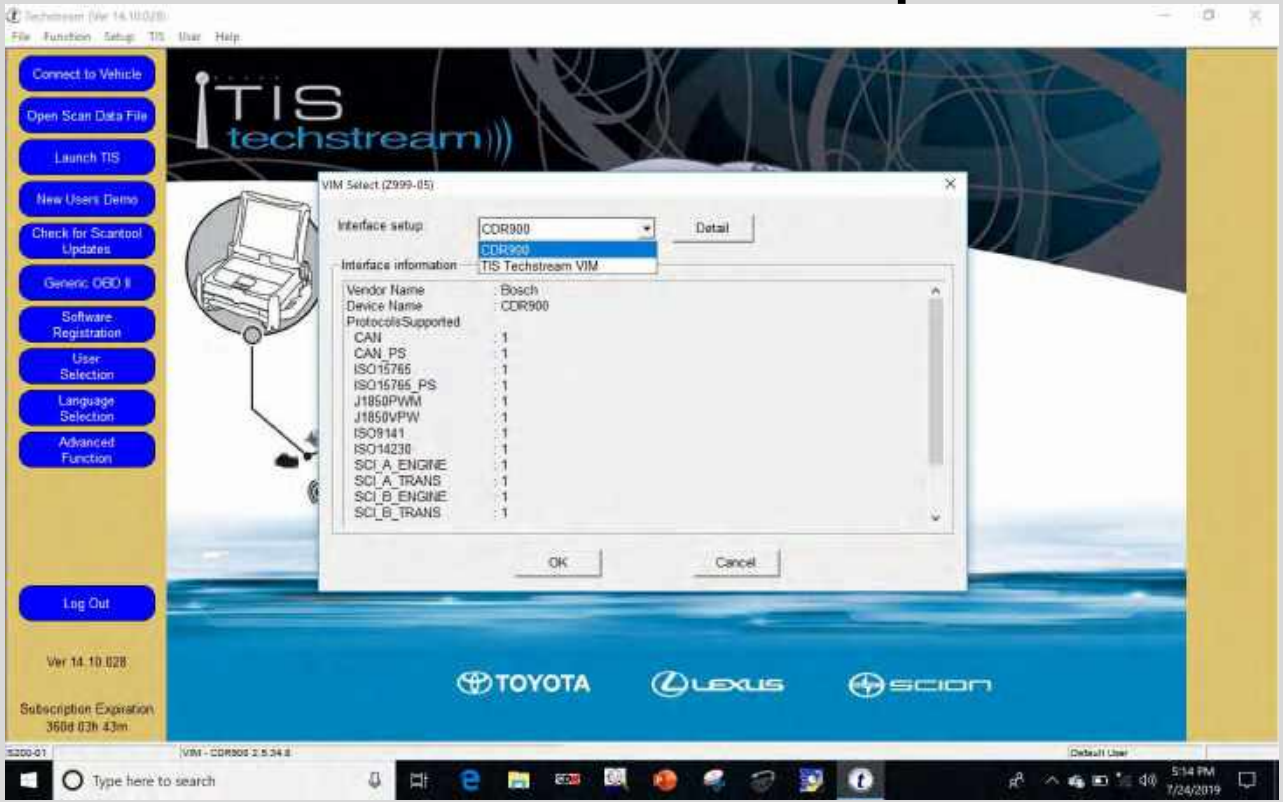
- Using CDR900 or other interface?
- Click “Setup”
- Click “VIM Select”



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Toyota TechStream VIM Setup

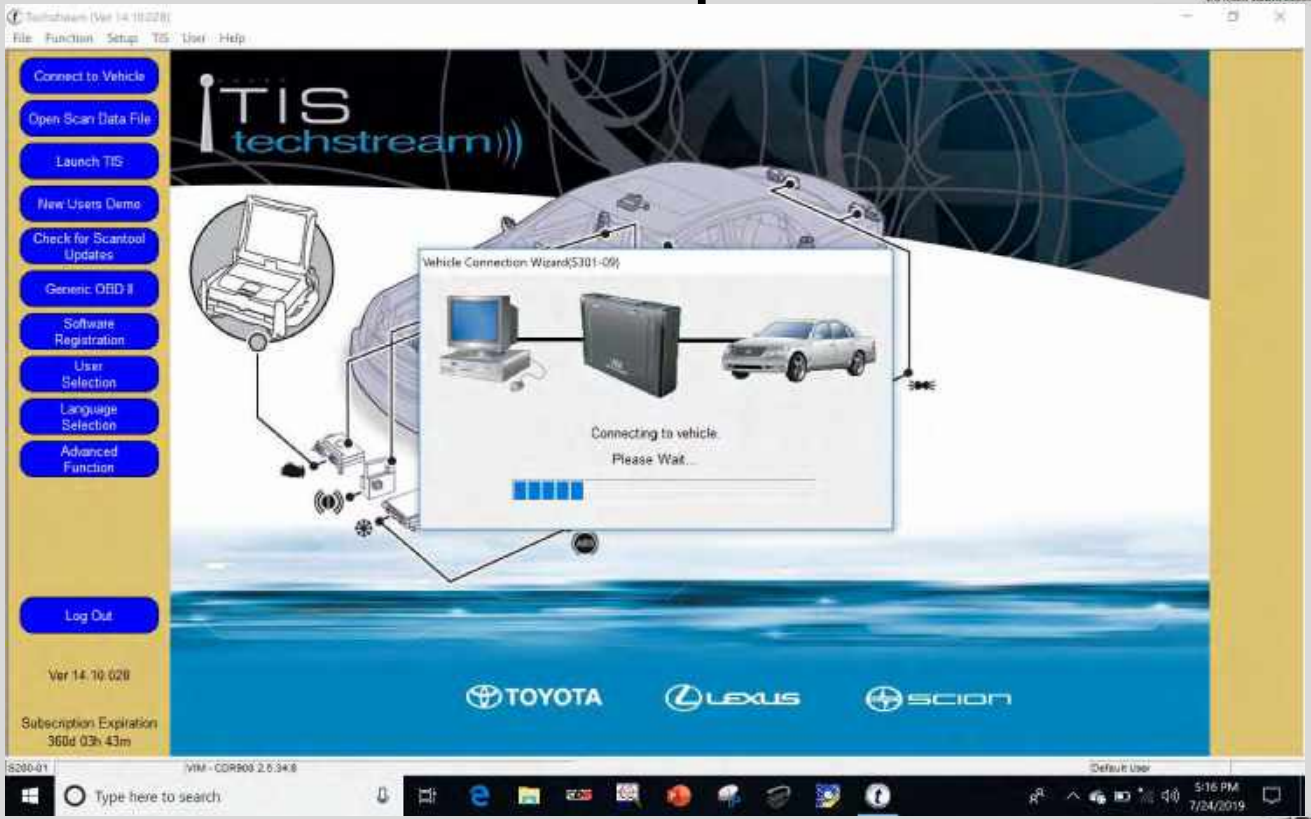
- Available “VIM”
  - V ehicle
  - I nterface
  - M odule
- Several may be in list
- Choose “CDR900”
- Click OK



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Toyota TechStream Operation

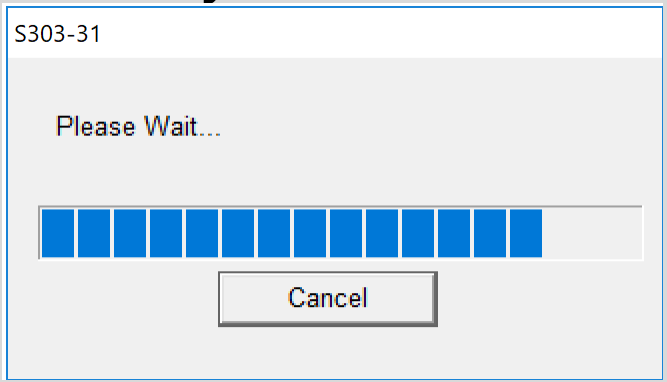
- Click “Connect to Vehicle”
- Make sure Ignition Key is ON !!

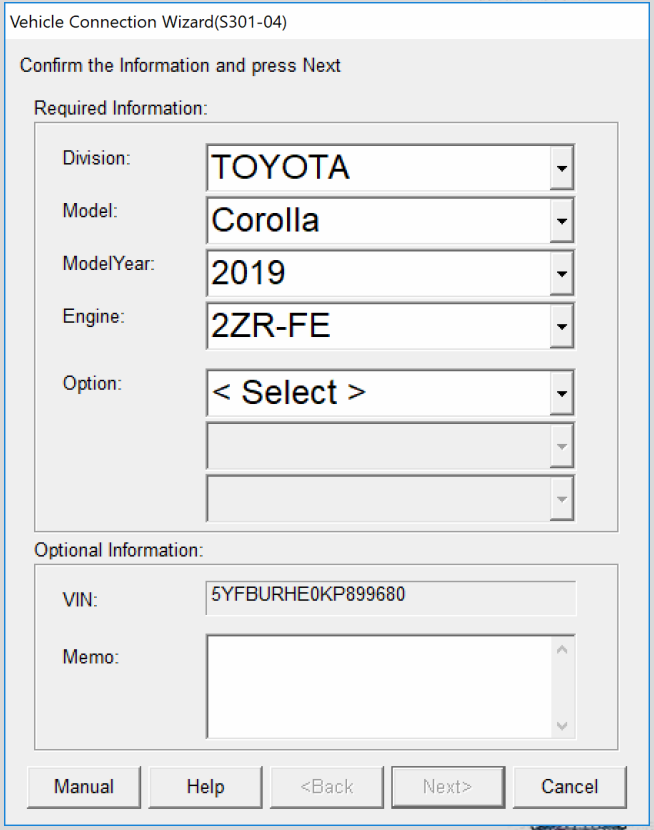


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Toyota TechStream Operation

- TechStream connects to the vehicle and should auto detect
- VIN auto populated
- Additional Options may need to be user selected:
  - Smartkey
  - Radar Cruise
- Memo area

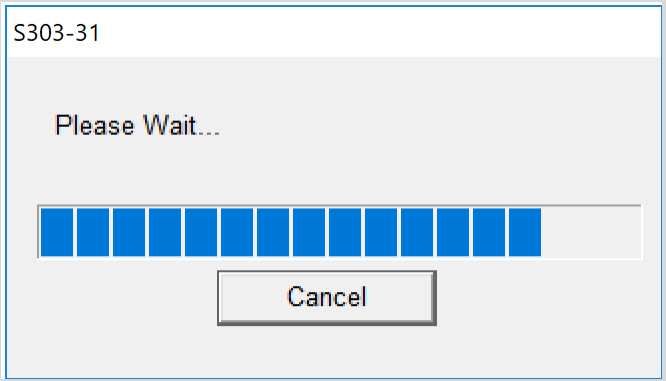


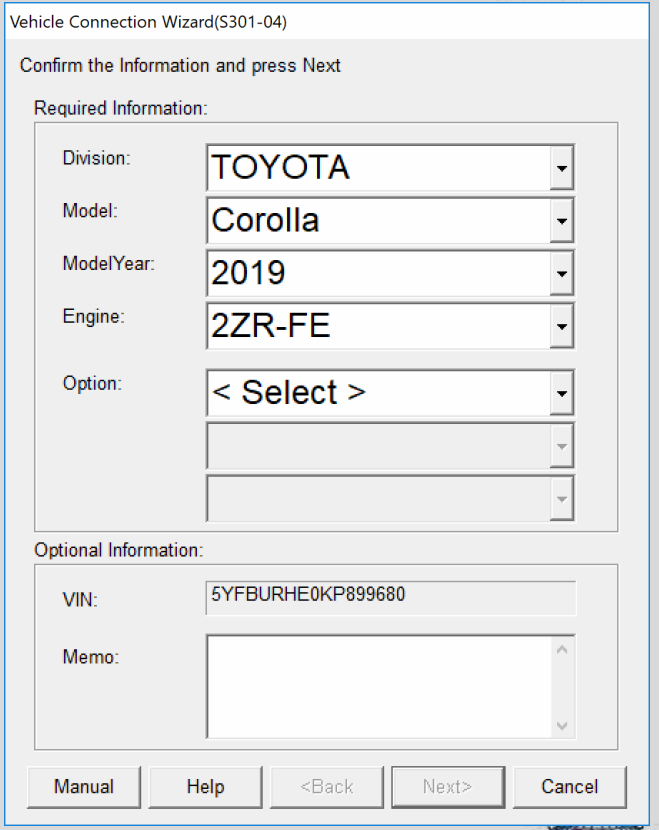


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Manual Connect – D2M With TechStream

- You will need to manually select Division / Model / Model Year / Engine / Options
- VIN MUST be entered manually!!
- Additional Options may need to be user selected:
  - Smartkey
  - Radar Cruise
- Memo area

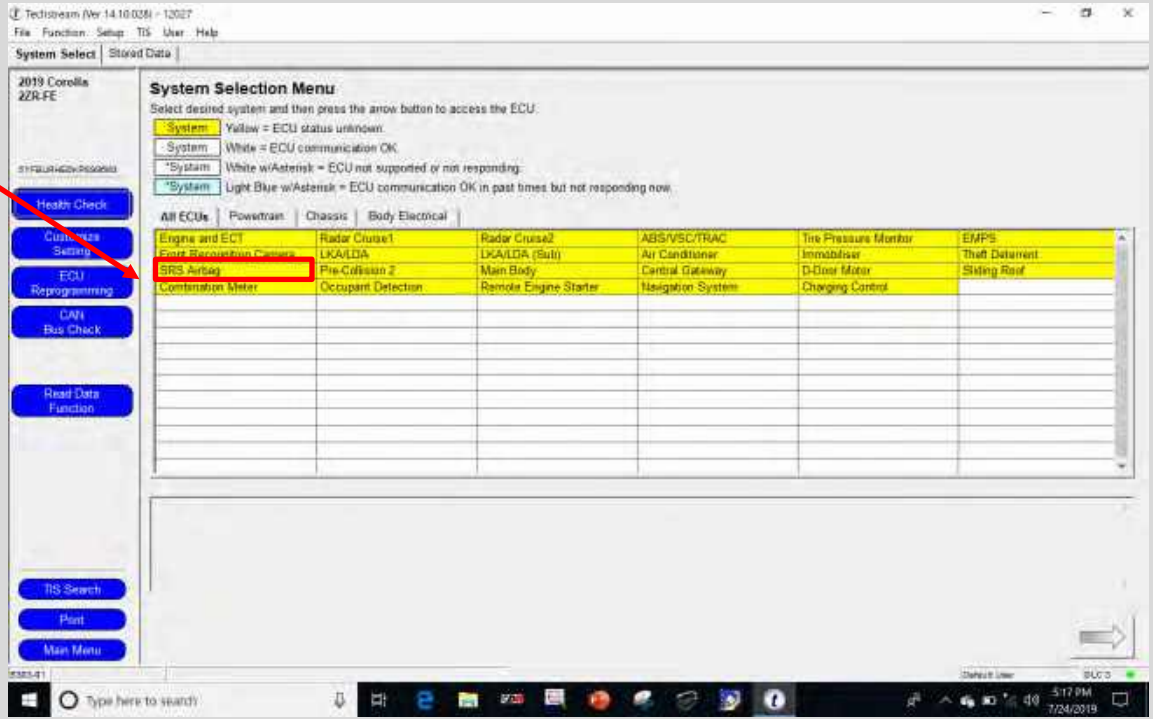




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DTC Viewing

- Click “SRS Airbag” or similar wording
- This allows for DTC display
- If connected to vehicle you can check DTC and “live” data for all ECU’s



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DTC Viewing

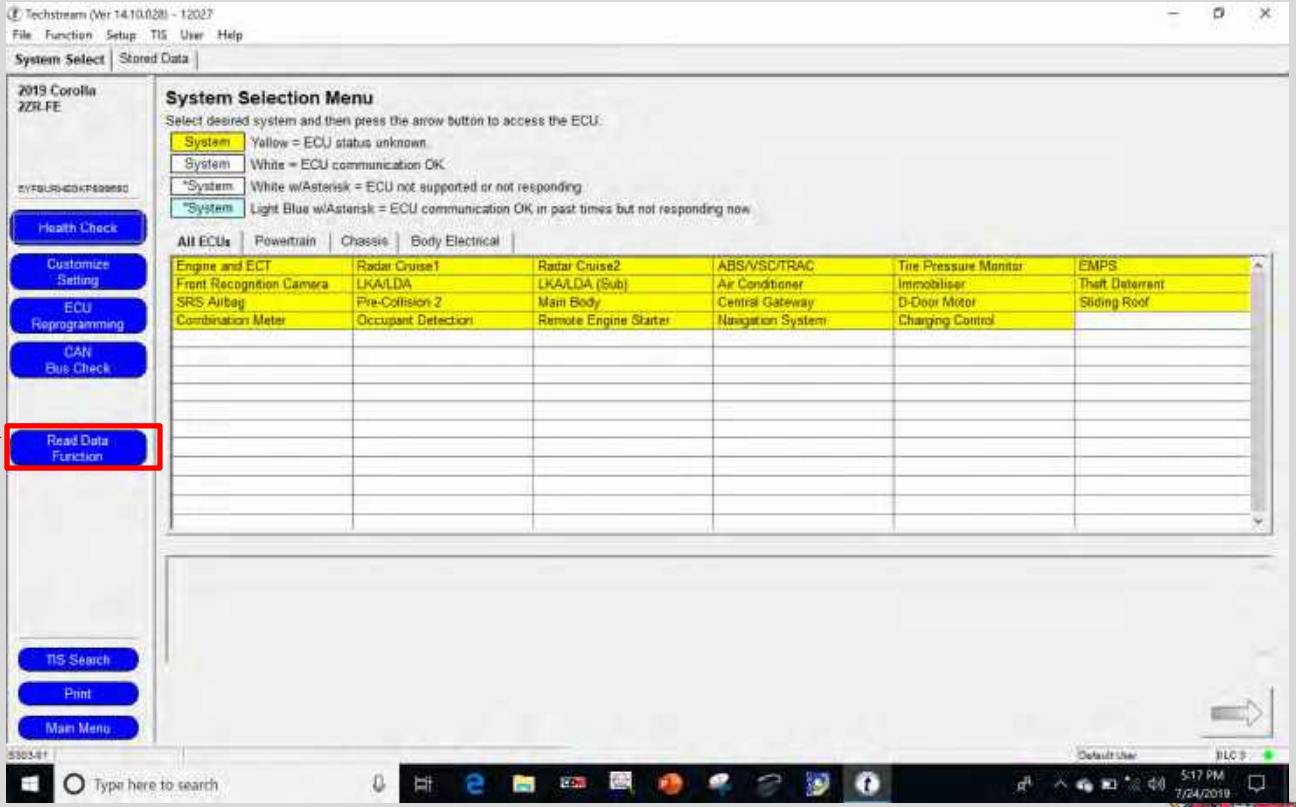
- After clicking “SRS Airbag” or similar
- Click “Trouble Codes” to view active DTC’s
- Here, it was a D2M connection



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Toyota TechStream Operation

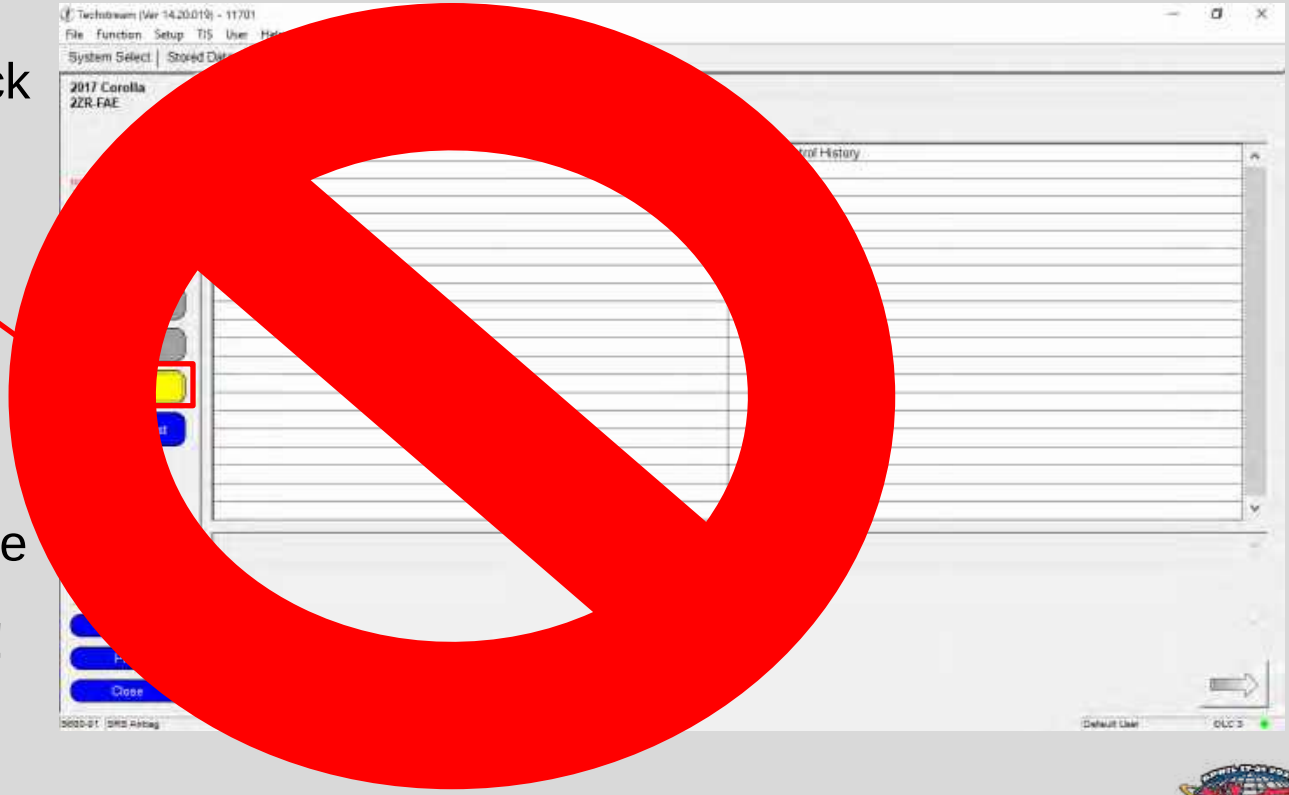
- Yellow ECU's are possible equipment
  - If you select "Health Check", after a Bus Scan, available ECU's show as White
- Click "Read Data Function"



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Alternate Route to VCH Retrieval

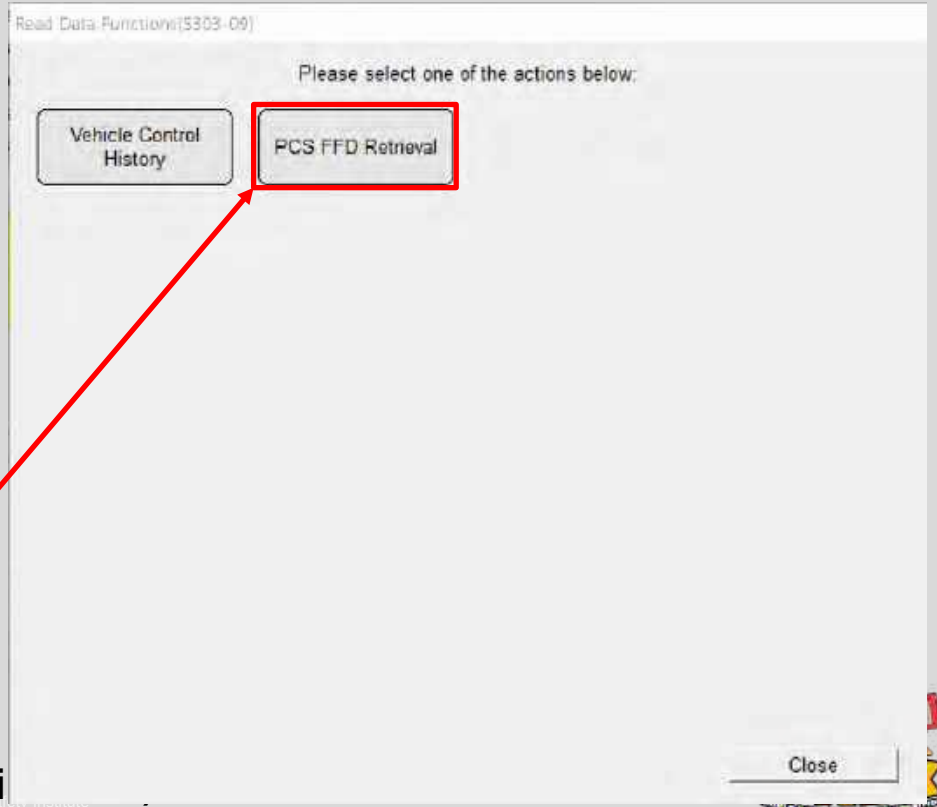
- While in “SRS Airbag” section, click “Utility”
- Clicking “Vehicle Control History” brings similar prompts to what follows
- NOT RECOMMENDED as this results in one large .CSV file and NO Image retrieval!



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Toyota TechStream Operation

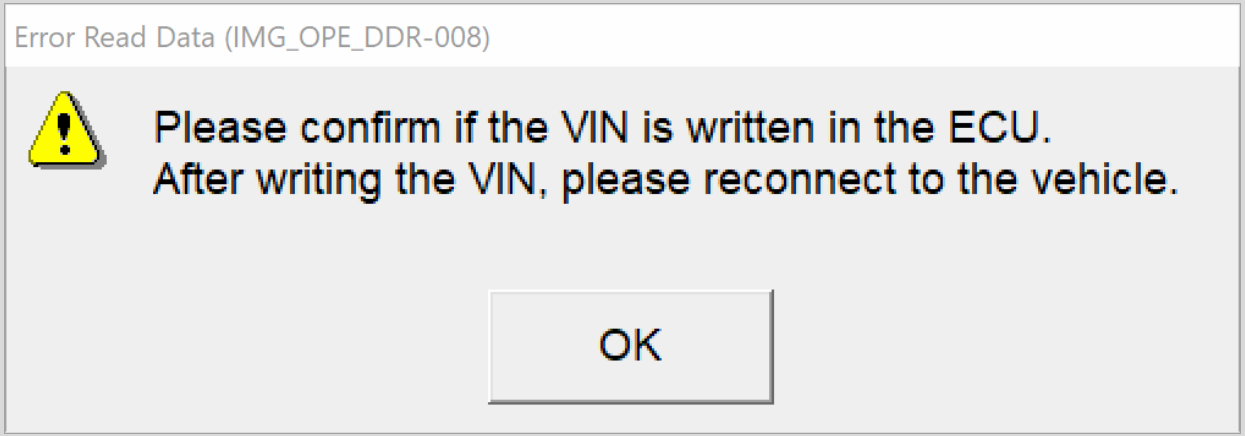
- Two Paths to the Data
  - Vehicle Control History
    - Only retrieves VCH
  - PCS FFD Retrieval
    - Retrieves VCH and Images (If present)
- Click “PCS FFD Retrieval”
  - (Vehicle Control History option results in one file with ALL data and no images)



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VIN Mismatch Error

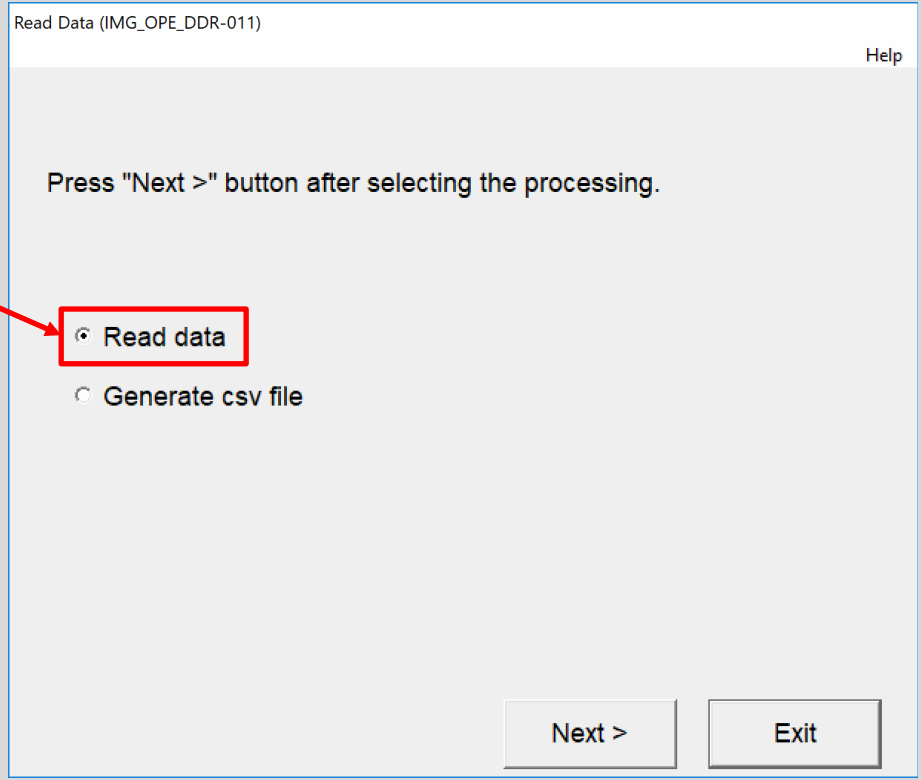
- When connecting manually you MUST input the VIN
- This error occurs since TechStream is looking to use the VIN in file creation



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Toyota TechStream Operation

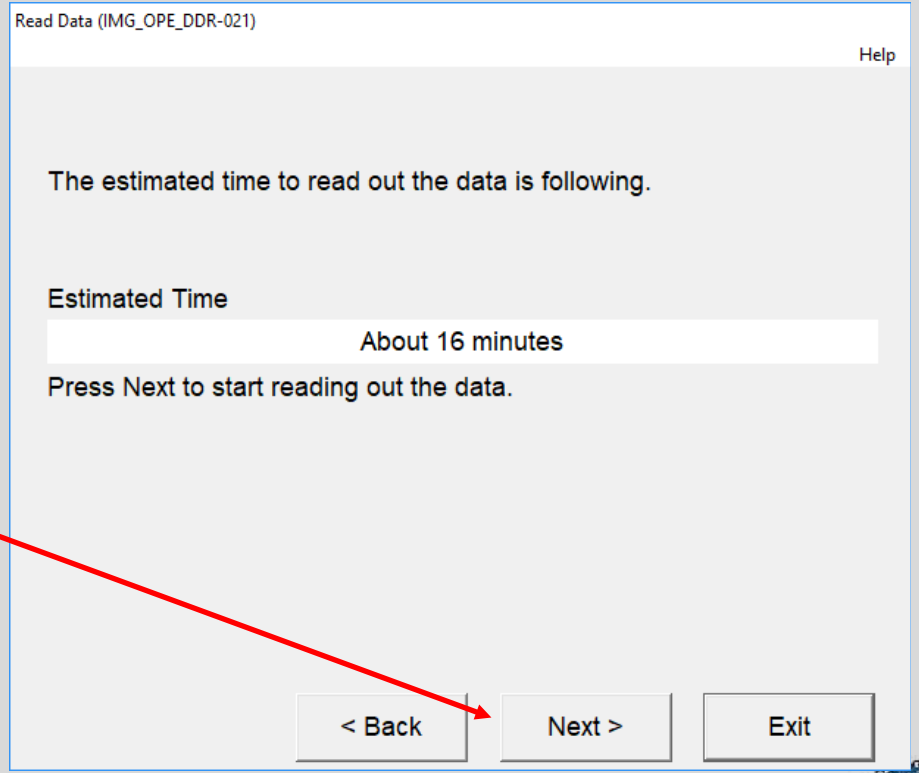
- Two Paths
  - Read Data
  - Generate csv file
- Click “Read data”
- Click Next
- (“Generate csv file” will not retrieve image data)



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Toyota TechStream TSS 2.0

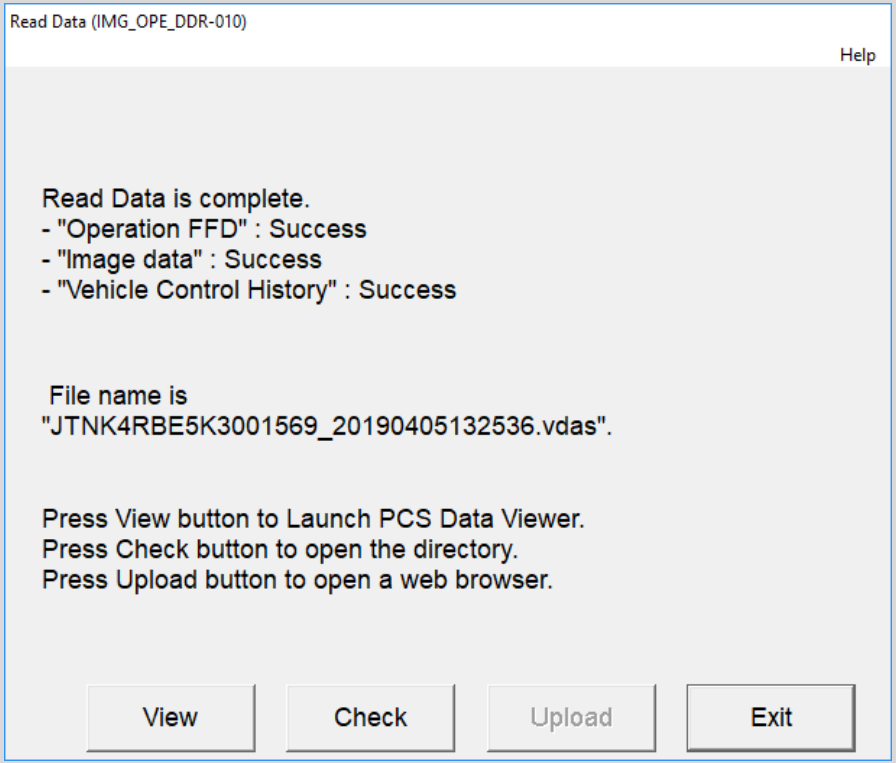
- Ensure sufficient battery power or booster battery available
- Click “Next” to begin reading the data



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Toyota TechStream TSS 2.0

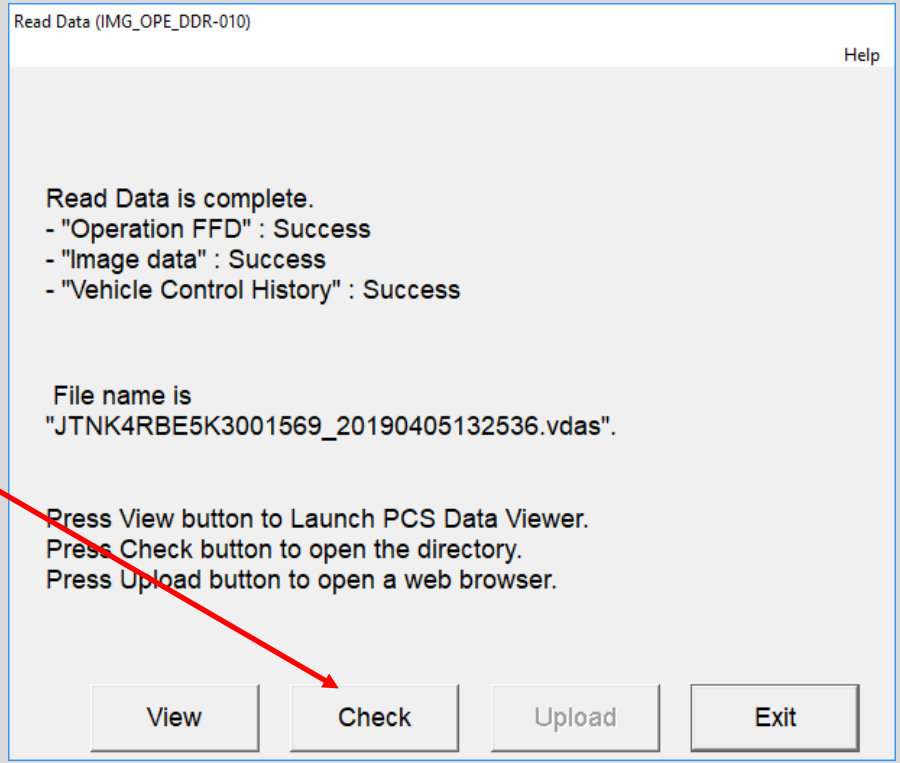
- Once complete, note the summary data displayed
- If “Image data” is N/A then no images were stored relating to a any events
- If “Vehicle Control History” is ”Success” then the VCH data has been retrieved



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Toyota TechStream VDAS File Viewing

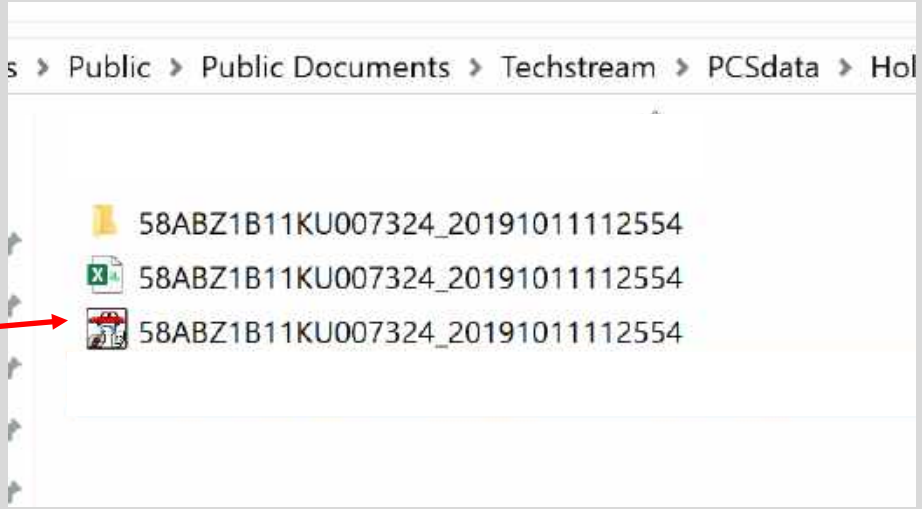
- Once complete, click “Check”
- This takes you to a File Explorer view of the location the .VDAS files are stored



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Toyota TechStream – VDAS Viewer

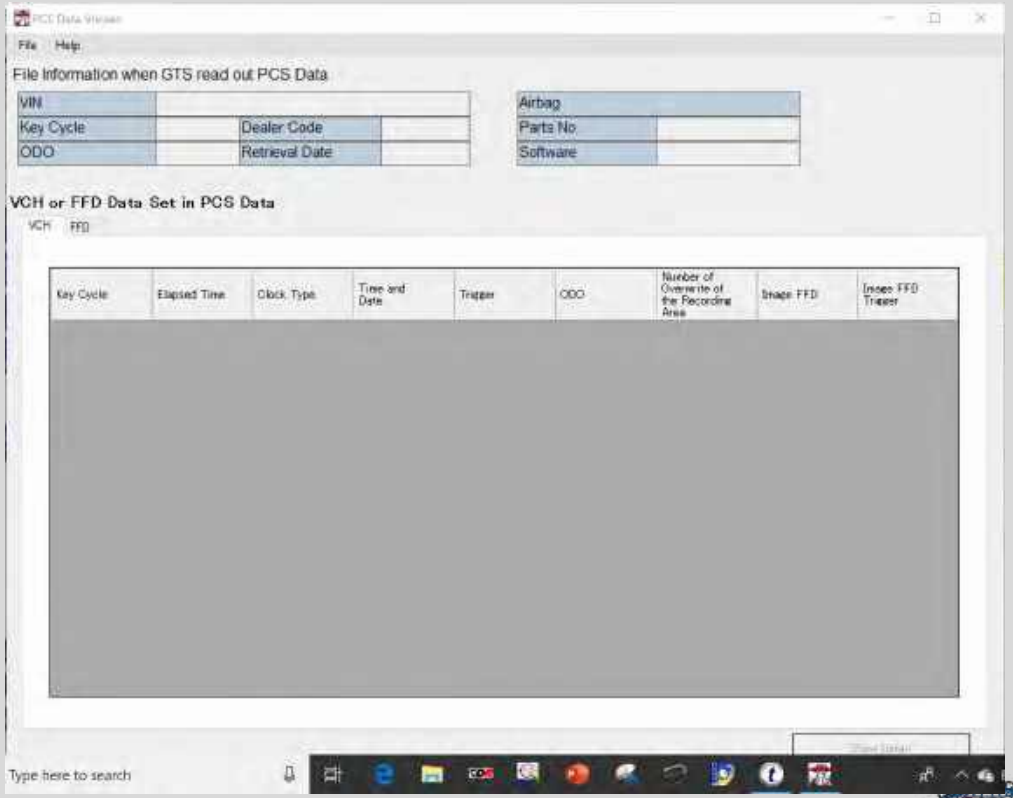
- After clicking “Check” a File Explorer view opens to the location the VDAS files are saved in.
- Double-clicking the .VDAS file will open it in the PCS Data Viewer



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Toyota TechStream – PCS Data Viewer

- Launch the PCS Data Viewer application
- Load the VDAS file saved in previous steps

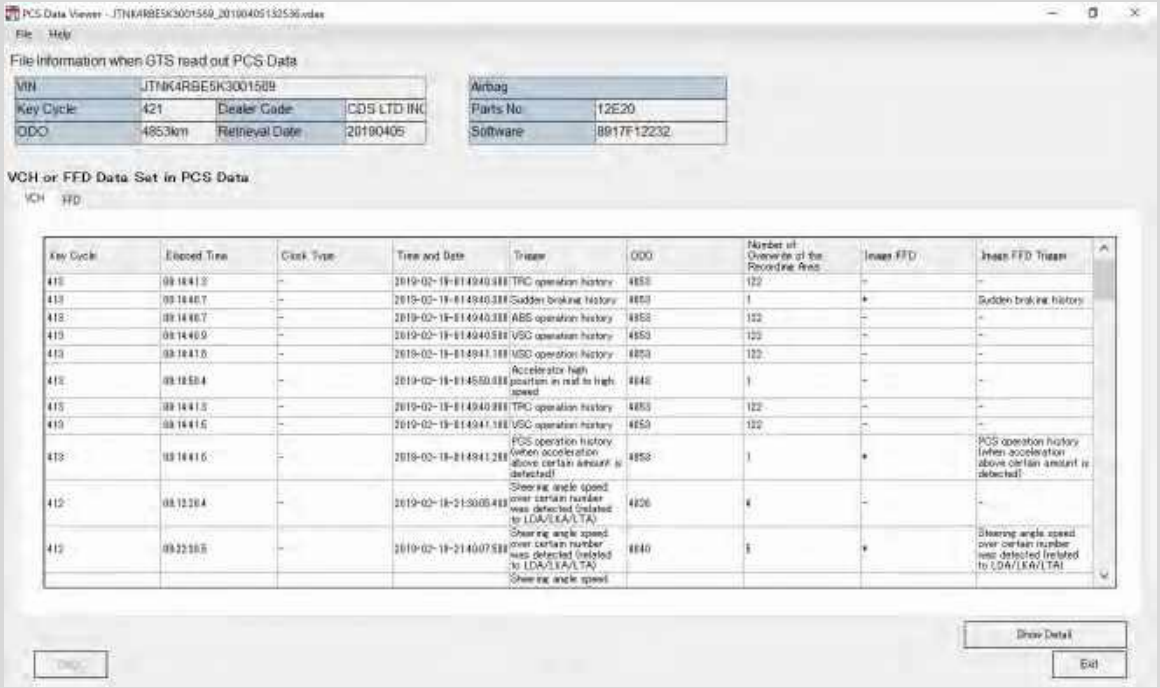


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Toyota TechStream – PCS Data Viewer

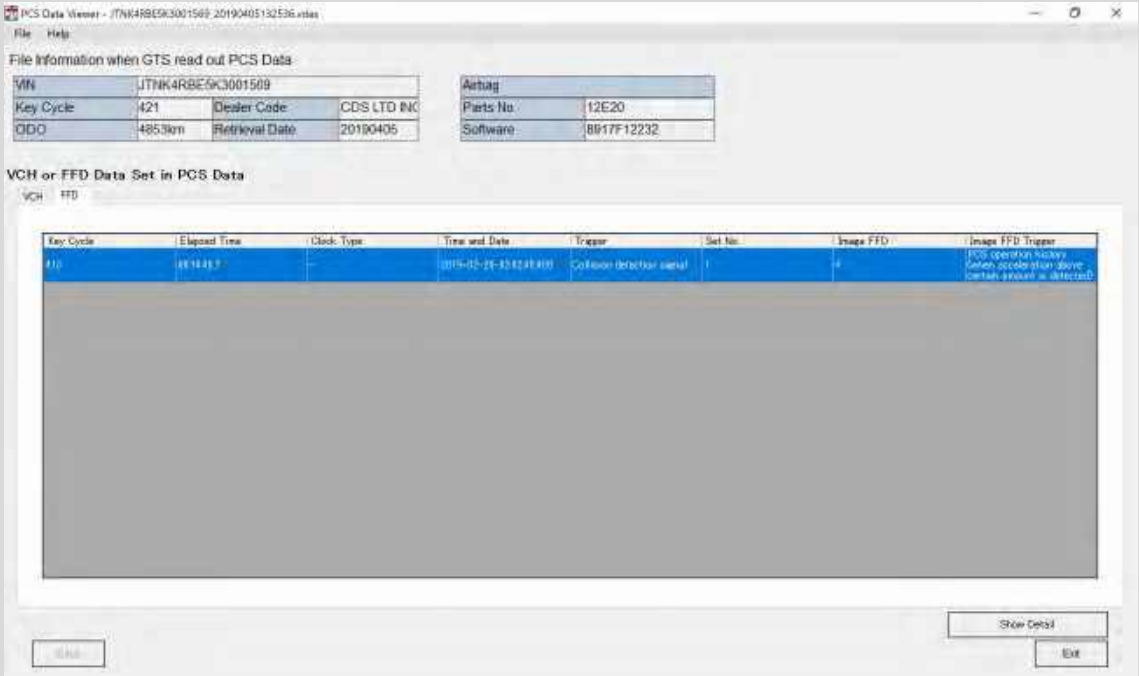
- Under the VCH Tab all VCH events are listed
- Double-Click “Key Cycle” to sort them from highest to lowest



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Toyota TechStream – PCS Data Viewer

- Under the FFD Tab “Freeze Frame Data” records are listed
- Double-Click “Key Cycle” to sort them from highest to lowest
- Events with an * in the Image FFD column will have associated images



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Toyota TechStream – PCS Data Viewer

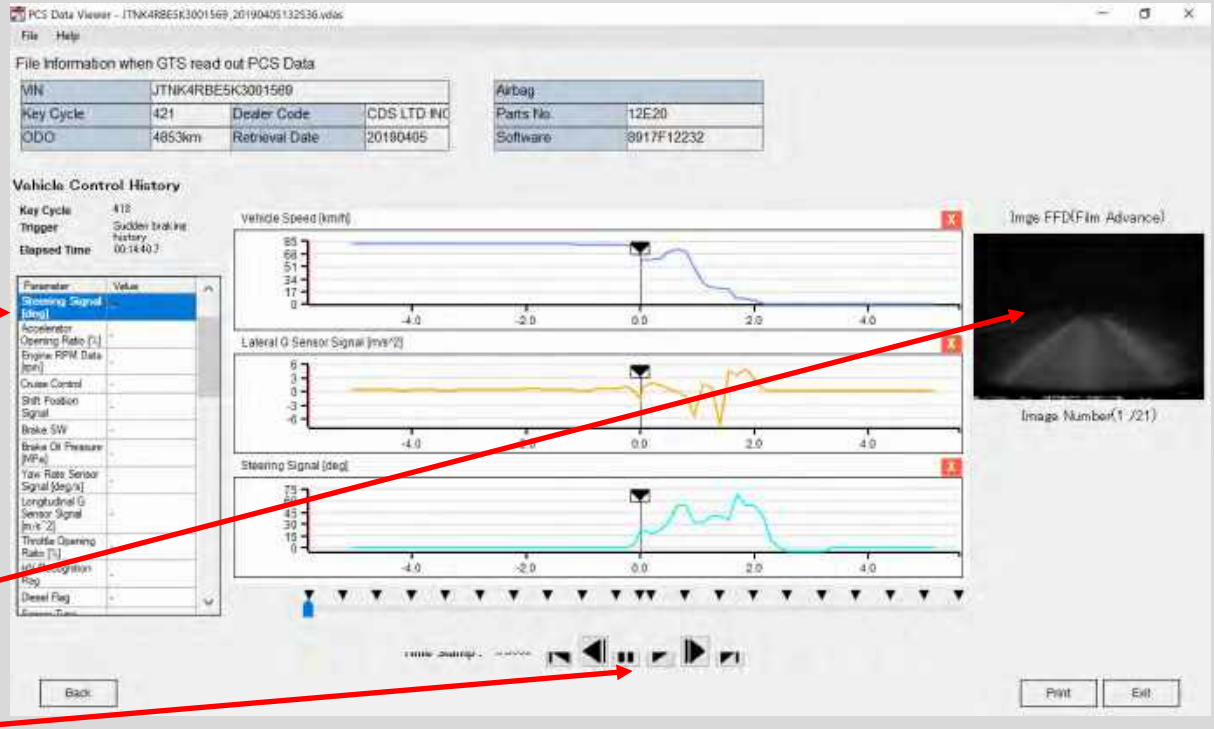
- VCH and FFD views of the same event contain both over-lapping and different data
- Double click an item to view playback
- Click the red “X” to remove



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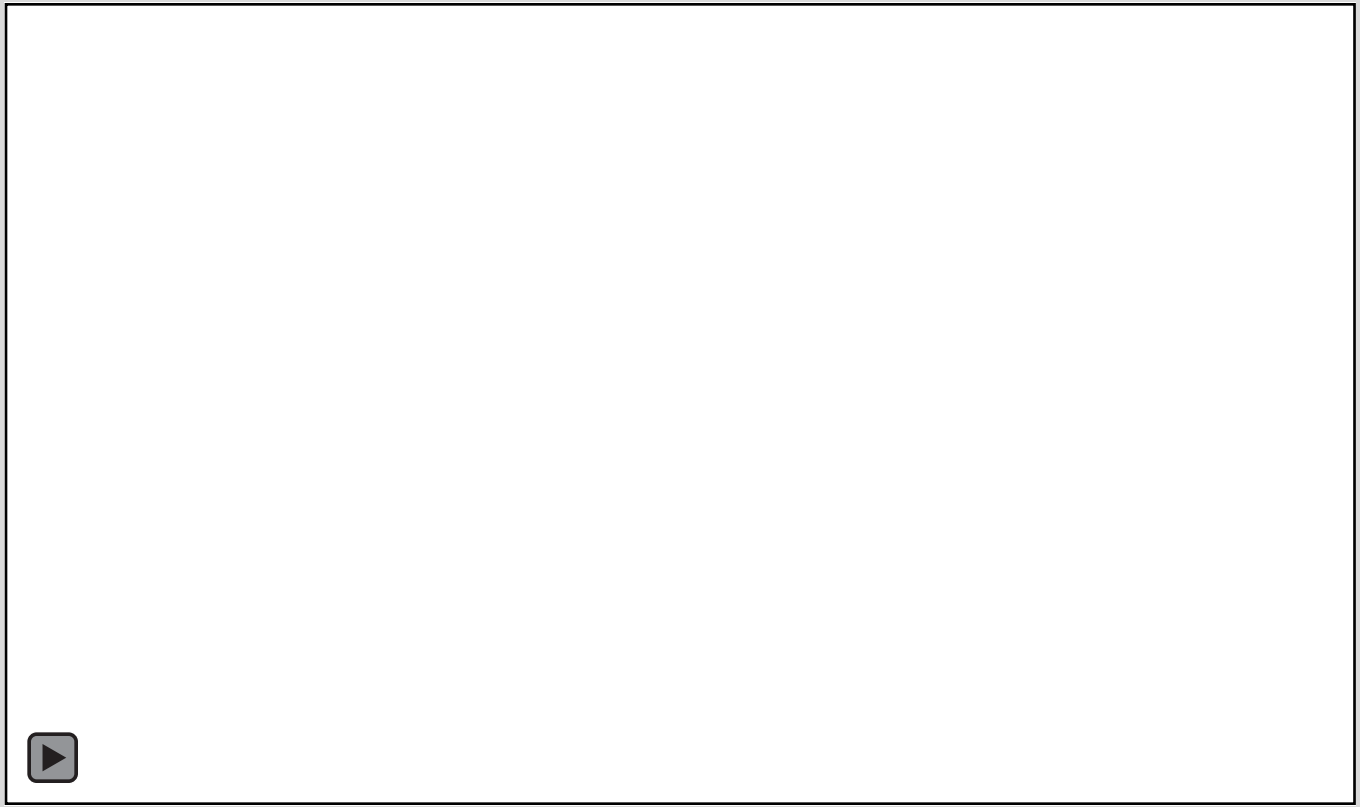
Toyota TechStream – PCS Data Viewer

- If either a VCH or FFD event with an * under Images is selected, and a parameter is selected for playback, associated images will be loaded for synched playback
- Playback viewer controls



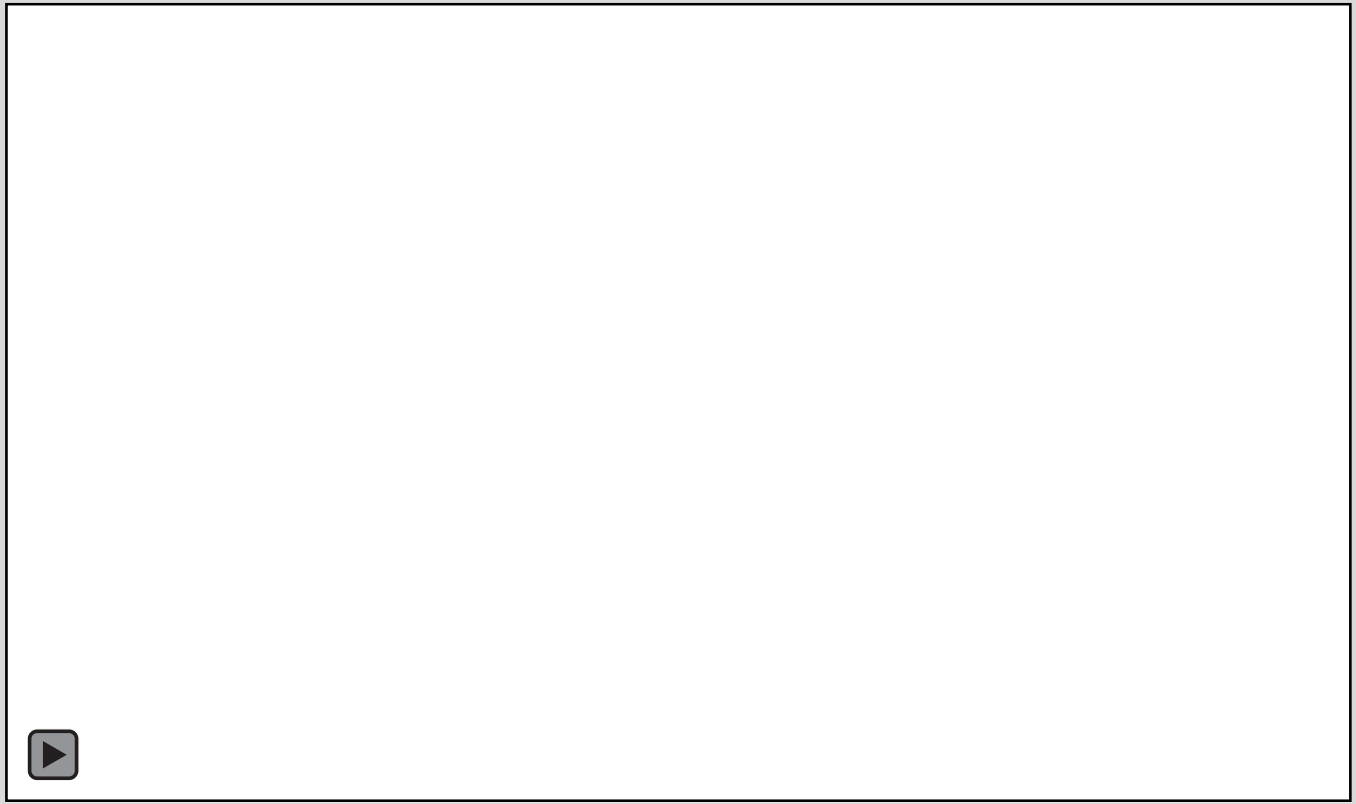
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Toyota TechStream – PCS Data Viewer



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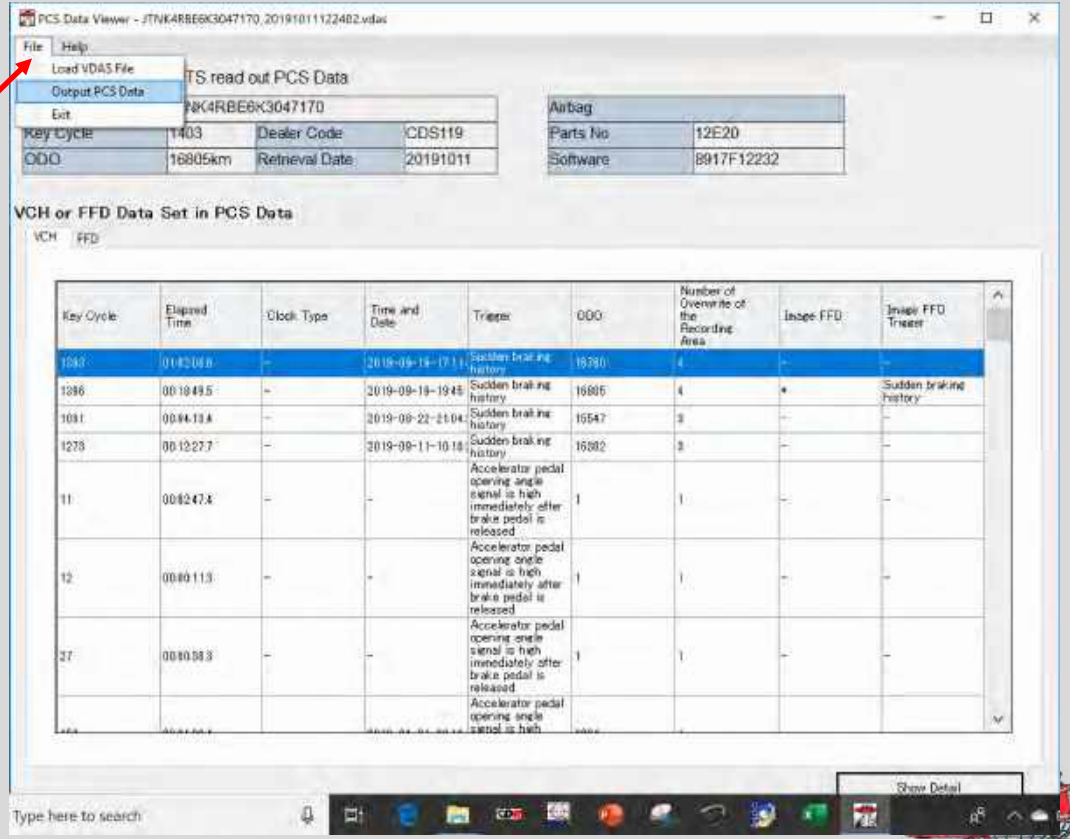
Toyota TechStream – PCS Data Viewer



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Toyota TechStream – PCS Data Output

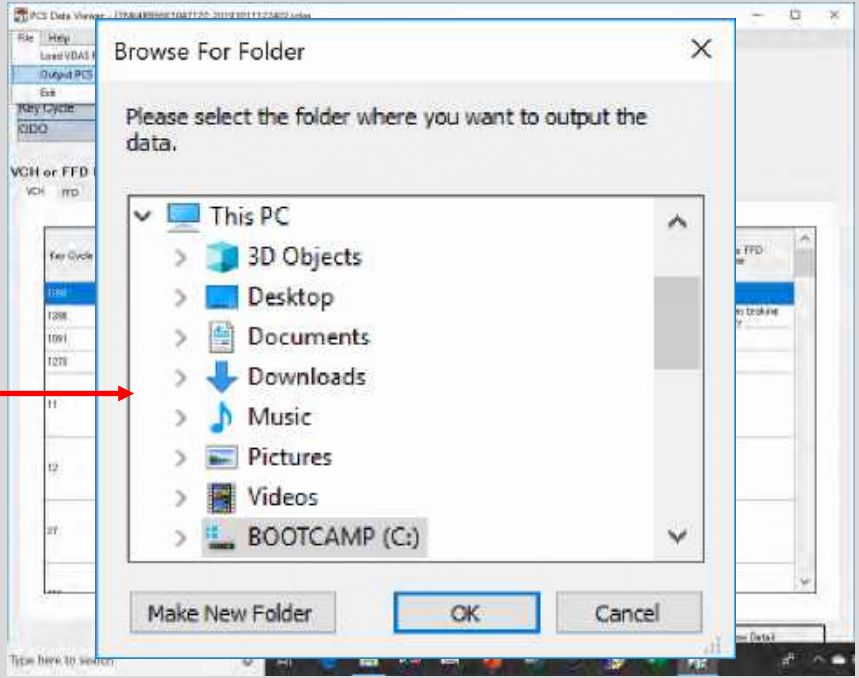
- To output the VCH or FFD data to event specific .CSV files and Images to separate image files:
- Click File / Output



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Toyota TechStream – PCS Data Output

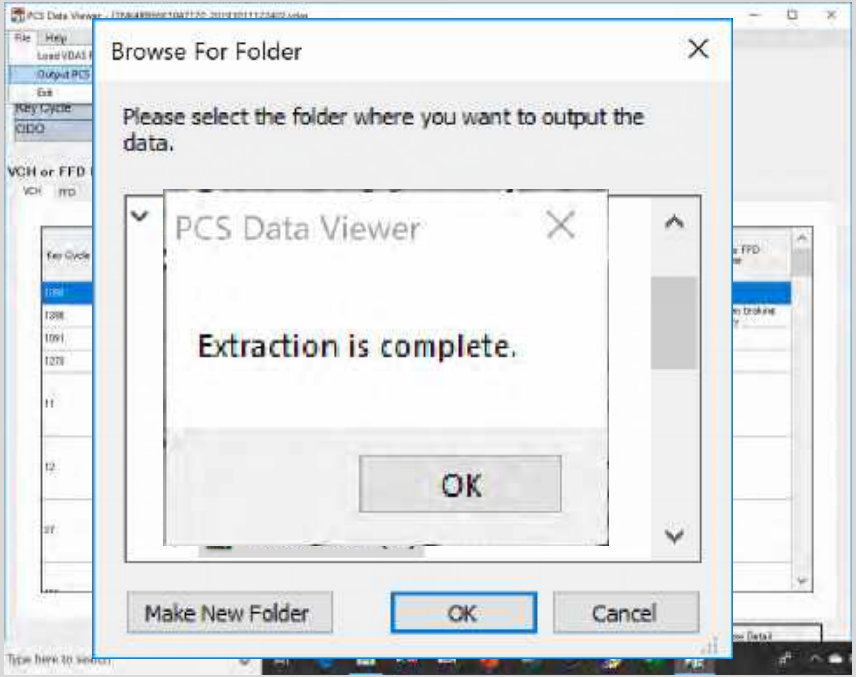
- To output the VCH or FFD data to event specific .CSV files and Images to separate image files:
- Click File / Output
- This you will open a Dialogue for you to chose where you want to save the output files



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Toyota TechStream – PCS Data Output

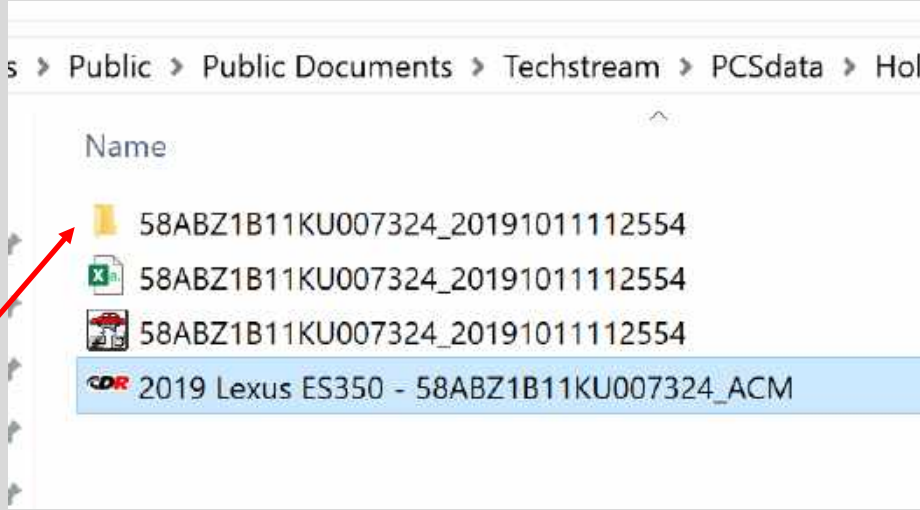
- To output the VCH or FFD data to event specific .CSV files and Images to separate image files:
- Click File / Output
- This you will open a Dialogue for you to chose where you want to save the output files
- Once Extraction is complete you can view the output files



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Toyota TechStream – PCS Data Output

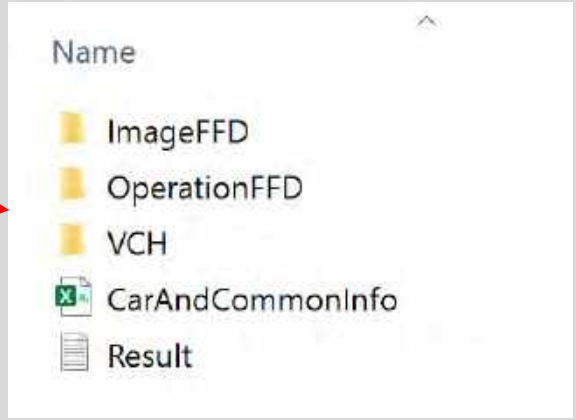
- Navigate to where you saved the output in the previous step.
- Here, it was saved in the same location as the .VDAS file and the .CDRX was also copied to this location
- Double-Click the VDAS output directory



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Toyota TechStream – PCS Data Output

- Output will be contained in 3 Directories
  - PCS Images
  - FFD .CSV data files
  - VCH .CSV files
- Car and Common info contains some basic vehicle and tool related data
- Result.txt displays which VCH areas have data

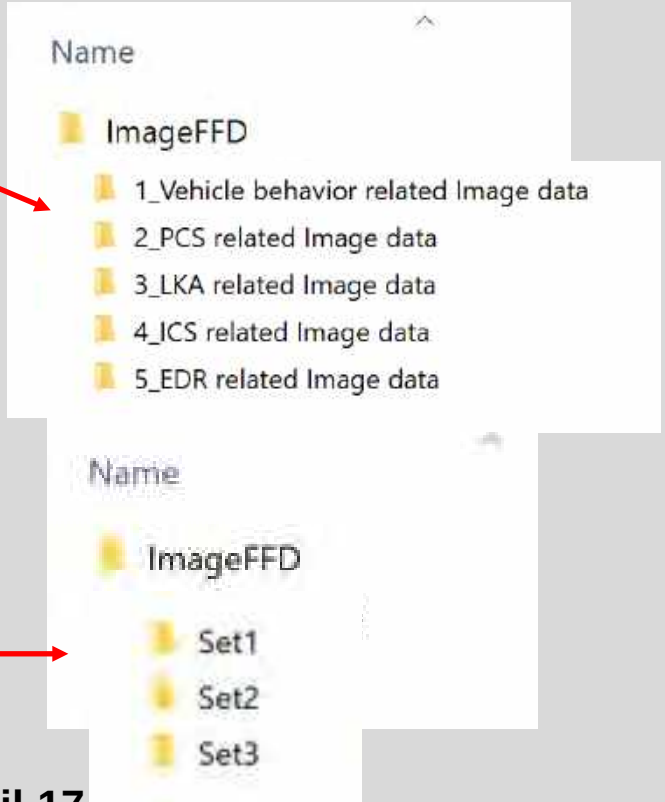


1	Car info	VIN	58ABZ1B11KU007324		
2		ECU info	Airbag	Parts No.	6880
3				software	8917F06451
4		Key Cycle	135		
5		Odo meter	163km		
6	Common inf	Dealer code	CDS LTD INC		
7		Retrieval Dat	20190405		

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Toyota TechStream – PCS Data Output

- For TSS => 2.0 supported vehicles, separate directories will contain images based on PCS trigger type
  - Images may be duplicated for concurrent triggers
- For TSS 1.0 there may be up to 3 sets if images only

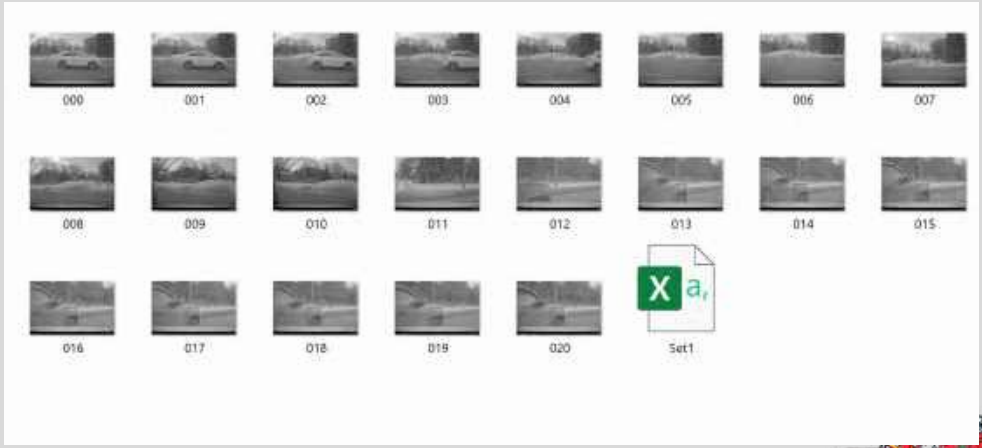


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Toyota TechStream – PCS Data Output

- Images
  - Each Category may contain from 1 to 3 sets of images
- Number if images before / after trigger depend on the event trigger type

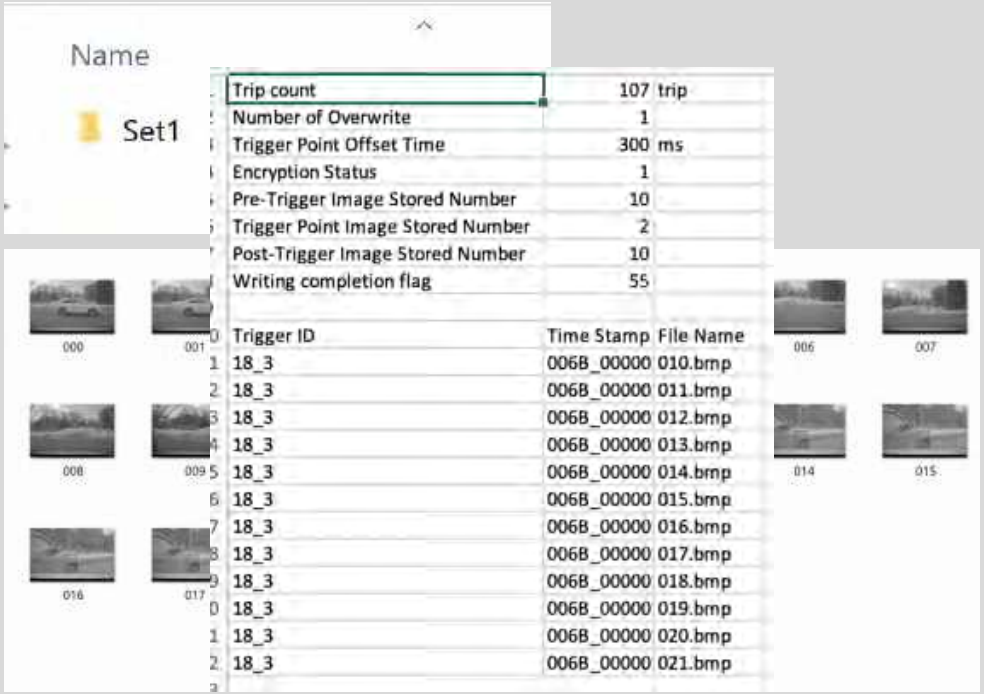




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Toyota TechStream – PCS Data Output

- Images
  - Each Category may contain from 1 to 3 sets of images
- Number if images before / after trigger depend on the event trigger type
- The “Setx” .CSV file contains information to assist in linking to CDR data for TSS 2.0
- TSS 1.0 does not create this file

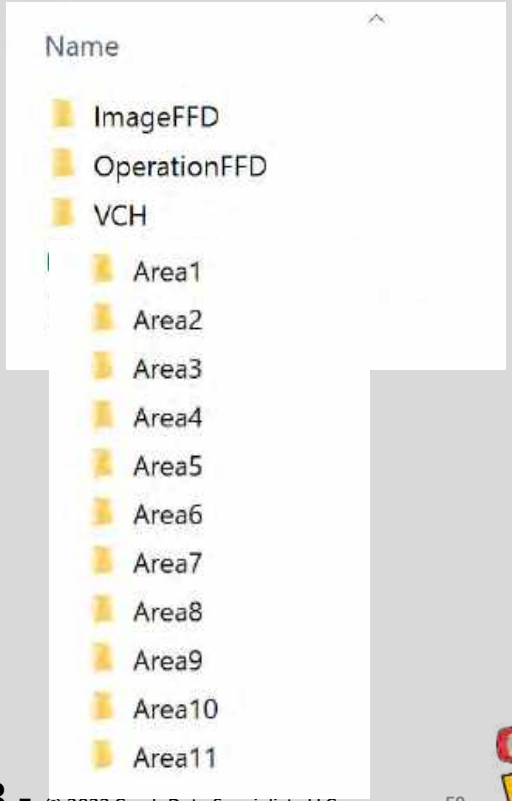


	Trip count	107	trip
	Number of Overwrite	1	
	Trigger Point Offset Time	300	ms
	Encryption Status	1	
	Pre-Trigger Image Stored Number	10	
	Trigger Point Image Stored Number	2	
	Post-Trigger Image Stored Number	10	
	Writing completion flag	55	
0	Trigger ID	Time Stamp	File Name
1	18_3	006B_00000	010.bmp
2	18_3	006B_00000	011.bmp
3	18_3	006B_00000	012.bmp
4	18_3	006B_00000	013.bmp
5	18_3	006B_00000	014.bmp
6	18_3	006B_00000	015.bmp
7	18_3	006B_00000	016.bmp
8	18_3	006B_00000	017.bmp
9	18_3	006B_00000	018.bmp
10	18_3	006B_00000	019.bmp
11	18_3	006B_00000	020.bmp
12	18_3	006B_00000	021.bmp

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Toyota TechStream – PCS Data Output

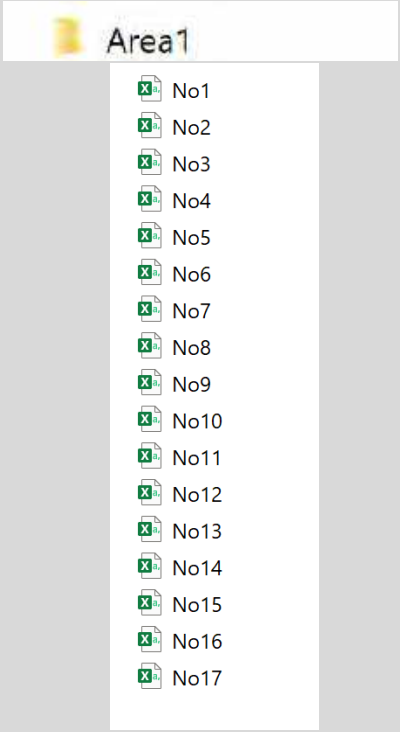
- Vehicle Control History events are sorted and written in “Areas”
- Each event type has an Area classification
- Each Classification has overwriting rules and a number of events that can be stored of that type



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Toyota TechStream – PCS Data Output

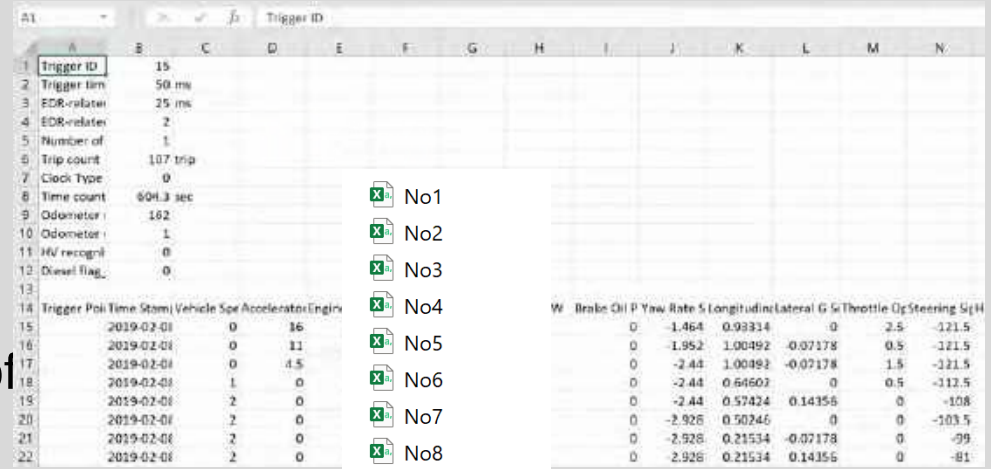
- Vehicle Control History events are sorted and written in “Areas”
- Each event type has an Area classification
- Each Classification has overwriting rules and a number of events that can be stored of that type
- Due to overwriting file No. may not be chronological
- .CSV Files written for each VCH Event Trigger



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Toyota TechStream – PCS Data Output

- Vehicle Control History events are sorted and written in “Areas”
- Each event type has an Area classification
- Each Classification has overwriting rules and a number of events that can be stored of that type
- Due to overwriting file No. may not be chronological



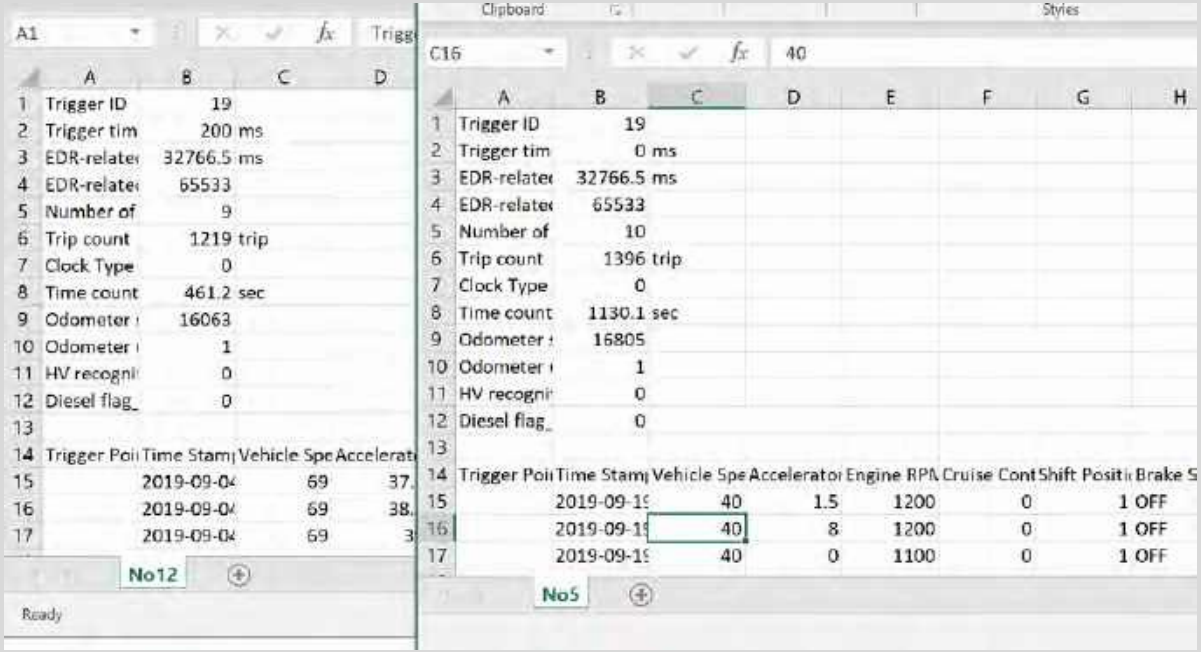
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Trigger ID	15												
2	Trigger tim	50 ms												
3	EDR-relater	25 ms												
4	EDR-relater	2												
5	Number of	1												
6	Trip count	107 trip												
7	Clock Type	0												
8	Time count	6041.3 sec												
9	Odometer	162												
10	Odometer	1												
11	HV recogni	0												
12	Diesel flag	0												
13														
14	Trigger Poi	Time Stam	Vehicle Spe	Accelerator	Engin				W	Brake Oil P	Yaw Rate S	Longitudin	Lateral G Si	Throttle Op	Steering Si	H
15		2019-02-08	0	16						0	-1.464	0.93314	0	2.5	-121.5	
16		2019-02-08	0	11						0	-1.952	1.00492	-0.07178	0.5	-121.5	
17		2019-02-08	0	4.5						0	-2.44	1.00492	-0.07178	1.5	-121.5	
18		2019-02-08	1	0						0	-2.44	0.64602	0	0.5	-112.5	
19		2019-02-08	2	0						0	-2.44	0.57424	0.14356	0	-108	
20		2019-02-08	2	0						0	-2.928	0.50246	0	0	-103.5	
21		2019-02-08	2	0						0	-2.928	0.21534	-0.07178	0	-99	
22		2019-02-08	2	0						0	-2.928	0.21534	0.14356	0	-81	

- No1
- No2
- No3
- No4
- No5
- No6
- No7
- No8
- No9
- No10
- No11
- No12
- No13
- No14
- No15
- No16
- No17

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Toyota TechStream – PCS Data Output

- Due to overwriting rules, the file No. may not be chronological!!
- Here, the Trip Count for Set No. 5 is HIGHER than the Trip Count for Set No. 12 indicating over-writes of this VCH data area



	A	B	C	D
1	Trigger ID	19		
2	Trigger tim	200 ms		
3	EDR-relate	32766.5 ms		
4	EDR-relate	65533		
5	Number of	9		
6	Trip count	1219 trip		
7	Clock Type	0		
8	Time count	461.2 sec		
9	Odometer	16063		
10	Odometer	1		
11	HV recogni	0		
12	Diesel flag_	0		
13				
14	Trigger Poi	Time Stam	Vehicle Spe	Accelerat
15		2019-09-0	69	37.
16		2019-09-0	69	38.
17		2019-09-0	69	3

No12

Ready

	A	B	C	D	E	F	G	H
1	Trigger ID	19						
2	Trigger tim	0 ms						
3	EDR-relate	32766.5 ms						
4	EDR-relate	65533						
5	Number of	10						
6	Trip count	1396 trip						
7	Clock Type	0						
8	Time count	1130.1 sec						
9	Odometer	16805						
10	Odometer	1						
11	HV recogni	0						
12	Diesel flag_	0						
13								
14	Trigger Poi	Time Stam	Vehicle Spe	Accelerato	Engine RPI	Cruise Cont	Shift Posit	Brake S
15		2019-09-1	40	1.5	1200	0	1 OFF	
16		2019-09-1	40	8	1200	0	1 OFF	
17		2019-09-1	40	0	1100	0	1 OFF	

No5

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Linking To a CDR Event

- Link “Trip Count”
- Link ”Time Count”
- (Available in 17EDR 19 EDR and newer)

System Status at Event (Most Recent Event, TRG 1)

TRG Count (times)	1
Event Type	Frontal/Rear/Side Crash
Previous Crash Type	No Event
Time from Previous TRG (msec)	32767 or greater
Time from Time Zero to TRG (msec)	84.0
Event Establishment Factor	Frontal Crash
TRG Establishment Factor	Frontal Crash
Freeze Signal	OFF
Freeze Signal Factor	None
Recording Status , Front/Rear and Side Crash Info.	Complete
Odometer signal (miles [km])	10,442 [16,805]
Trip count (times)	1396
Time count (msec)	1,130,100
Time count input system	Normal

	A	B	C
1	Trigger ID	19	
2	Trigger tim	0	ms
3	EDR-relate	32766.5	ms
4	EDR-relate	65533	
5	Number of	10	
6	Trip count	1396	trip
7	Clock Type	0	
8	Time count	1130.1	sec
9	Odometer :	16805	
10	Odometer	1	
11	HV recogni	0	
12	Diesel flag_	0	
13			

14	Trigger Point	Time Stamp	Vehicle Speed [km/h]	Accelerator Opening Ratio [%]	Engine RPM Data [rpm]	Cruise Control	Shift Position Signal	Brake SW	Brake Oil Pressure [MPa]	Yaw Rate Sensor Signal [deg/s]	Longitudinal G Sensor Signal [m/s^2]	Lateral G Sensor Signal [m/s^2]	Throttle Opening Ratio [%]	Steering Signal [deg]	HV Re n
15		2019-09-19	40	1.5	1200	0	1	OFF	0	-1.464	-0.28712	0.14356	0	0	
16		2019-09-19	40	8	1200	0	1	OFF	0	-1.464	0.21534	0.50246	2.5	0	

Toyota TechStream – PCS Data Printing

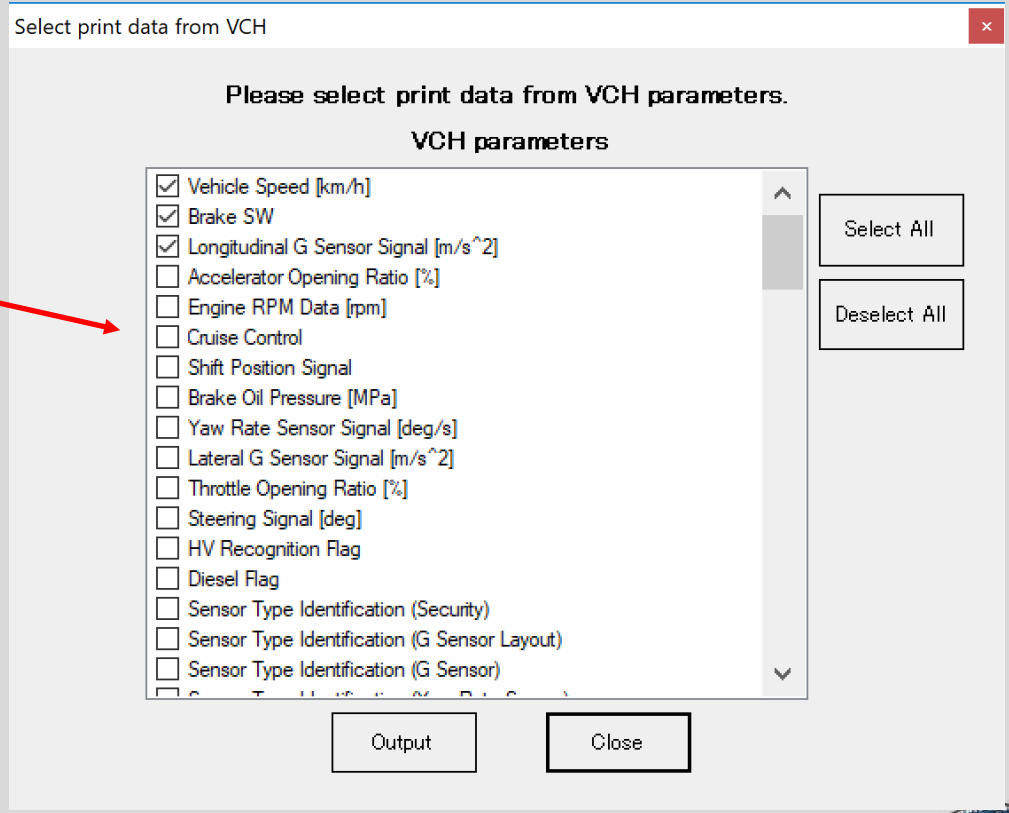
- Output can be printed to generate a full report
- Click “Print”
- You do NOT need to have items in the viewing area



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Toyota TechStream – PCS Data Printing

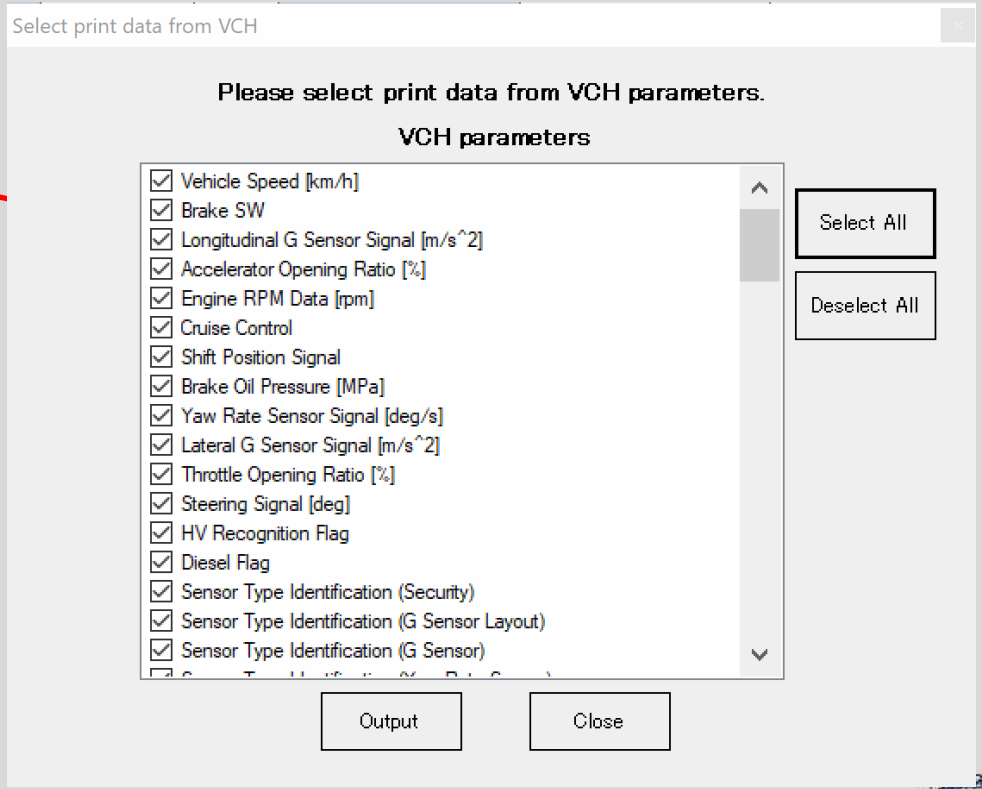
- You can select all, or some of the parameters to print
- Note: Images will be printed at bottom of report



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Toyota TechStream – PCS Data Printing

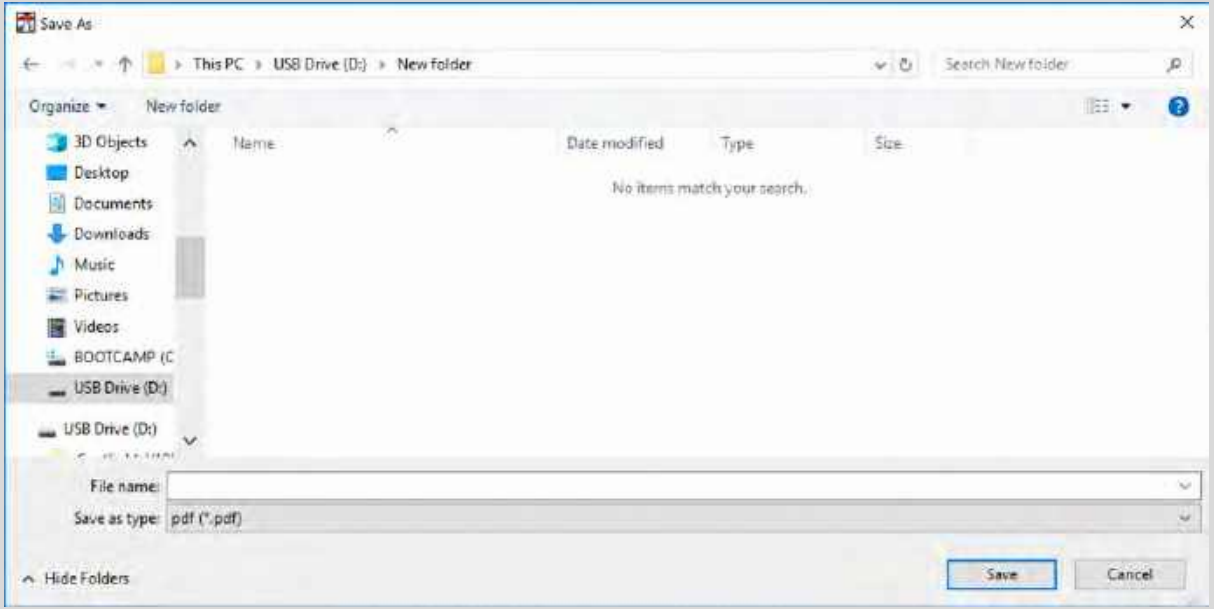
- You can select all, or some of the parameters to print
- Note: Images will be printed at bottom of report



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Toyota TechStream – PCS Data Printing

- You can select all, or some of the parameters to print
- Note: Images will be printed at bottom of report
- File Explorer window opens, navigate to Save location of choice



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Toyota TechStream – PCS Data Printing

PCS Data Report

This report created by PCS Data Viewer.
PCS Data Viewer data output

1. General information
2. File information
3. User selected VCH data and graph
4. Image FFD
5. Hexadecimal Data

1.General Information

1.1.Data Limitations

- Due to limitations of the data recorded, such as the resolution, data range, sampling interval, time period of the recording, and the items recorded, the information provided by this data may not be sufficient to capture the entire event.
- The recorded data is recorded in discrete intervals.

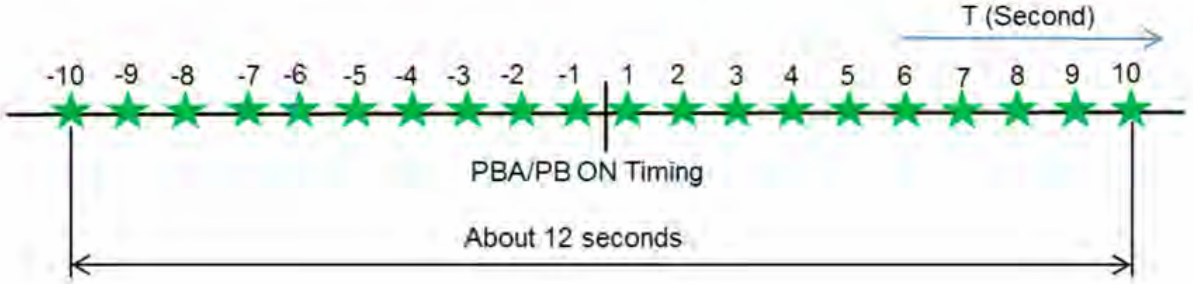
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Toyota TechStream – PCS Data Printing

- The following PCS data (Operation FFD) is recorded upon the PCS operation.
  - Vehicle velocity
  - Accelerator pedal position
  - Brake pedal(ON/OFF)
  - The operation situation of each function of PCS
  - Information such as the distance with objects such as preceding vehicles and their relative velocity
- PCS data (Image FFD) is recorded upon the pre-collision brake operation (PBA, PB).
  - In addition to the above, system records image information (Image FFD) of the camera sensor.
- Image information, Image FFD, is recorded 20 shots×3 events in TSS-P/LSS+, 2 shots×3events in TSS-C and 20 shots×10 events in LSS+A, and also maximum 10 shots at every trigger including the first trigger each event if another trigger occurs after the first trigger for six seconds.

Image FFD
- TSS-P/LSS+ PCS : 20shots
- LSS+A PCS : 20 shots+ Maximum 10 shots at every trigger including the first trigger

a) Pattern A



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Toyota TechStream – PCS Data Printing

1.3 Appendix : The meaning of Abbreviation

Abbreviation	Meaning
ABS	Antilock Brake System
ACC	Adaptive Cruise Control
AEB	Autonomous Emergency Braking
AHB	Automatic High-Beam
AIM	AIM (Camera axis)
ALM	ALarM
ARS	Adaptive Radar Sensor
ASL	Adaptive Speed Limiter
AVS	Adaptive Variable Suspension
CAA	Collision Avoidance Assist

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Toyota TechStream – PCS Data Printing

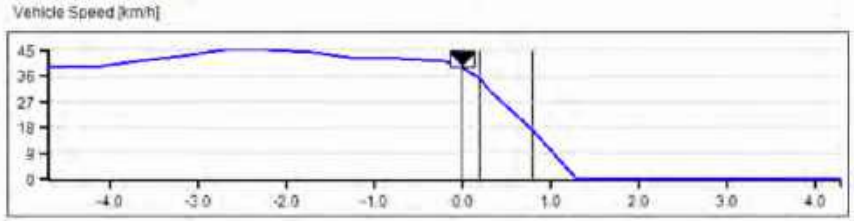
3.User Selected VCH or FFD Data

Trigger Name:PCS operation history (warning buzzer operation)

Elapsed Time: 00:18:49.3

3.1 Vehicle Speed [km/h]

Time Stamp	Parameter Value	Time Stamp	Parameter Value
-4.700	39		
-4.200	39		
-3.700	41		
-3.200	43		
-2.700	45		
-2.200	45		
-1.700	44		
-1.200	42		
-0.700	42		
-0.200	41		
0.000	39		
0.000	39		
0.200	35		
0.200	35		
0.300	31		
0.800	17		
0.800	17		
1.300	0		
1.800	0		
2.300	0		
2.800	0		
3.300	0		
3.800	0		
4.300	0		



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Toyota TechStream – PCS Data Printing

4.Image FFD

-5.400s



-3.000s



-4.800s



-2.400s



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Toyota Peripheral Camera Images

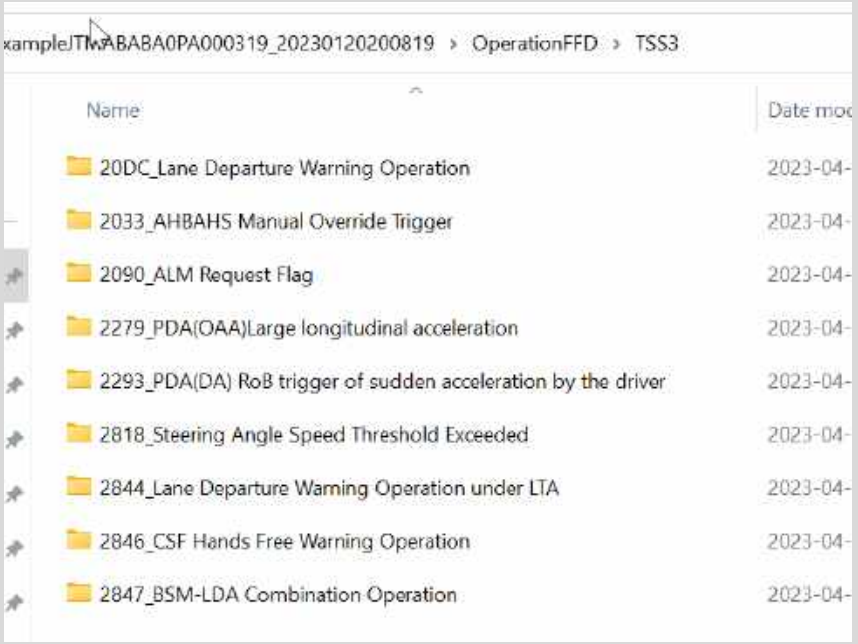
- BZ4X / Lexus NX
- TSS 3.0
- VDAS file must be exported
- Look in Export Directory Image “Set” directories



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Toyota Peripheral Camera Images

- Different Layout of Output Data
- Operation FFD Directories now labeled with event type
- Only way to see peripheral camera images
  - These images do NOT display in current VDAS viewer



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Toyota Peripheral Camera Images

- No Summary Block at top of CSV file with Trip Count / Time / Odo
- This data is now in a column within the data

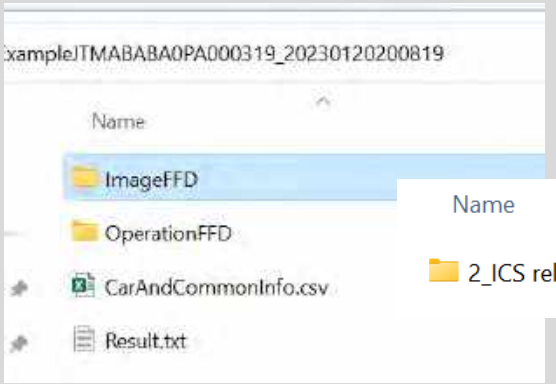
N8 : 87.9

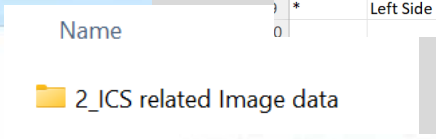
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Trigger Point	RoBCode	Odometer signal	Odometer unit signal	Time Infor	Time Infor	Time Infor	Time Infor	Time Infor	Time Infor	Time Stamp	Trip count [trip]	Time Stamp (Time Counter) [ms]	Additional Time counter [sec]
2		2279	35	km	22	8	30	13	57	7	2022-08-30-13:57:07.000	297	86200	87.2
3		2279	35	km	22	8	30	13	57	7	2022-08-30-13:57:07.200	297	86300	87.4
4		2279	35	km	22	8	30	13	57	7	2022-08-30-13:57:07.300	297	86400	87.5
5		2279	35	km	22	8	30	13	57	7	2022-08-30-13:57:07.400	297	86500	87.6
6		2279	35	km	22	8	30	13	57	7	2022-08-30-13:57:07.500	297	86600	87.7
7		2279	35	km	22	8	30	13	57	7	2022-08-30-13:57:07.500	297	86700	87.7
8		2279	35	km	22	8	30	13	57	7	2022-08-30-13:57:07.700	297	86800	87.9

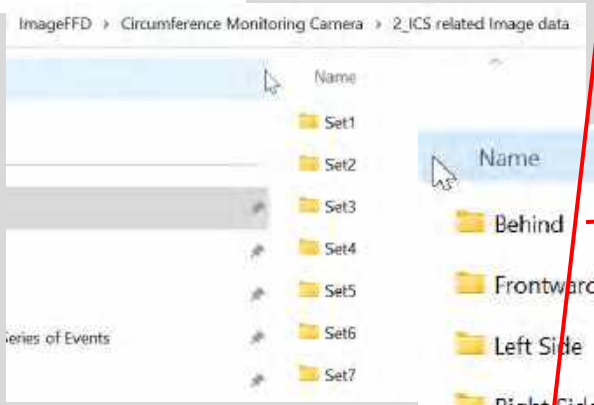
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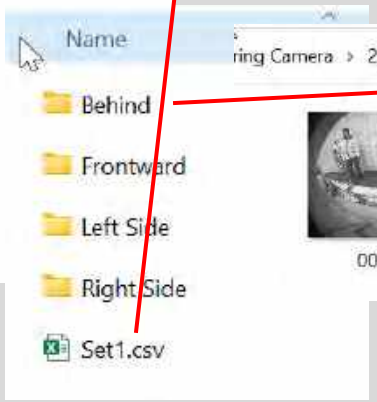
Toyota Peripheral Camera Images

Trigger Po	Camera direction	RoBCode	Time Stamp	File Name	Trigger ID	Trip count [trip]	Time Stamp (Time Counter) [ms]	Odometer unit signal
	Frontward	2825	2022-11-17-08:57:04.000	000.bmp	20_3	410	3890	km
*	Frontward	2825	2022-11-17-08:57:05.000	001.bmp	20_3	410	4890	km
	Behind	2853	2022-11-17-08:57:04.000	000.bmp	20_3	410	3890	km
*	Behind	2853	2022-11-17-08:57:05.000	001.bmp	20_3	410	4890	km
	Right Side	2855	2022-11-17-08:57:04.000	000.bmp	20_3	410	3890	km
*	Right Side	2855	2022-11-17-08:57:05.000	001.bmp	20_3	410	4890	km
	Left Side	2854	2022-11-17-08:57:04.000	000.bmp	20_3	410	3890	km
*	Left Side	2854	2022-11-17-08:57:05.000	001.bmp	20_3	410	4890	km











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Toyota Peripheral Camera Images

ng Camera › 2_ICS related Image data › Set1 › Frontward


000.bmp


001.bmp

ng Camera › 2_ICS related Image data › Set1 › Left Side


000.bmp


001.bmp

ng Camera › 2_ICS related Image data › Set1 › Right Side


000.bmp


001.bmp

ring Camera › 2_ICS related Image data › Set1 › Behind


000.bmp


001.bmp

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LKA / ICS / EDR Related



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EDR (CDR) vs. VCH Data (#19 – Set 10)

System Status at Event (Most Recent Event, TRG 3)

TRG Count (times)	3
Event Type	Rollover
Previous Crash Type	Frontal/Rear/Side Crash
Time from Previous TRG (msec)	37.5
Freeze Signal	OFF
Freeze Signal Factor	None
Recording Status, Rollover Crash Info.	Complete
Odometer signal (miles [km])	101 [162]
Trip count (times)	107
Time count	604.400
Time count input system	Normal

Trigger ID	19	
Trigger time information	200	ms
EDR-related information	12.5	ms
EDR-related information	3	
Number of Overwrite of the Recording Area	1	
Trip count	107	trip
Clock Type	0	
Time count	604.4	sec
Odometer signal	162	
Odometer unit signal	1	
HV recognition flag_header	0	
Diesel flag_header	0	

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CDR Data

Pre-Crash Data -5 to 0 Seconds (Most Recent Event, TRG 3) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-4.80	0.0 [0]	0.0	0.5	Invalid	900	Invalid	OFF
-4.30	1.2 [2]	0.0	0.0	Invalid	600	Invalid	OFF
-3.80	1.9 [3]	1.5	0.0	Invalid	600	Invalid	OFF
-3.30	1.9 [3]	19.5	5.0	Invalid	700	Invalid	OFF
-2.80	3.7 [6]	40.0	9.0	Invalid	1,300	Invalid	OFF
-2.30	5.6 [9]	40.0	9.0	Invalid	1,600	Invalid	OFF
-1.80	7.5 [12]	41.0	11.5	Invalid	1,900	Invalid	OFF
-1.30	9.3 [15]	39.5	12.0	Invalid	2,000	Invalid	OFF
-0.80	11.2 [18]	44.0	20.5	Invalid	2,500	Invalid	OFF
-0.30	10.6 [17]	0.0	0.0	Invalid	1,200	Invalid	ON
TRG(0)	5.6 [9]	0.0	0.0	Invalid	900	Invalid	ON

Pre-Crash Data -5 to 0 Seconds (Most Recent Event, TRG 3) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-4.80	OFF	OFF	0.00	0.861	-2.44	-120.0	D
-4.30	OFF	OFF	0.00	0.215	-2.44	-102.0	D
-3.80	OFF	OFF	0.00	0.359	-2.44	-67.5	D
-3.30	OFF	OFF	0.00	0.359	-2.44	-52.5	D
-2.80	OFF	OFF	0.00	1.866	-1.46	-31.5	D
-2.30	OFF	OFF	0.00	-0.431	-1.95	-15.0	D
-1.80	OFF	OFF	0.00	1.507	0.00	18.0	D
-1.30	OFF	OFF	0.00	2.010	12.20	142.5	D
-0.80	OFF	OFF	0.00	1.651	30.74	265.5	D
-0.30	OFF	OFF	4.94	-4.737	-0.98	-33.0	D
TRG(0)	ON	OFF	10.56	-8.973	-61.00	-166.5	D

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Corresponding VCH Data

Time Stamp	Vehicle Speed [km/h]	Accelerator Opening Ratio [%]	Engine RPM Data [rpm]	Brake SW	Brake Oil Pressure [MPa]	Yaw Rate Sensor Signal [deg/s]	Longitudinal G Sensor Signal [m/s^2]	Lateral G Sensor Signal [m/s^2]	Throttle Opening Ratio [%]	Steering Signal [deg]
2019-02-08-16:02:22.400	0	0	600	ON	2.592	-0.976	-0.07178	0.07178	0	-103.5
2019-02-08-16:02:22.800	0	0	500	ON	2.544	-0.976	-0.07178	0.07178	0	-103.5
2019-02-08-16:02:23.200	0	0	600	ON	1.008	-0.976	-0.07178	0.07178	0	-103.5
2019-02-08-16:02:23.600	0	0	600	ON	0	-0.976	-0.07178	0.07178	0	-112.5
2019-02-08-16:02:24.000	0	0	600	OFF	0	-0.976	-0.07178	0.07178	0	-112.5
2019-02-08-16:02:24.400	0	0	600	OFF	0	-0.976	-0.07178	0.07178	0	-121.5
2019-02-08-16:02:24.800	0	14	700	OFF	0	-0.976	0	0.07178	3	-121.5
2019-02-08-16:02:25.200	0	15	900	OFF	0	-1.464	0.93314	-0.07178	2.5	-121.5
2019-02-08-16:02:25.600	1	0	800	OFF	0	-2.44	0.64602	0	0.5	-112.5
2019-02-08-16:02:26.000	2	0	600	OFF	0	-2.928	0.21534	-0.07178	0	-99
2019-02-08-16:02:26.400	3	0	600	OFF	0	-2.44	0.28712	0	0	-67.5
2019-02-08-16:02:26.800	3	11	600	OFF	0	-2.44	0.14356	0.21534	3.5	-49.5
2019-02-08-16:02:27.200	4	38.5	1100	OFF	0	-2.44	1.36382	0.07178	6	-40.5
2019-02-08-16:02:27.600	7	40	1400	OFF	0	-3.416	1.86628	0.14356	8	-31.5
2019-02-08-16:02:28.000	10	40	1600	OFF	0	-1.464	1.22026	0.86136	9	-13.5
2019-02-08-16:02:28.400	12	41	1800	OFF	0	0	1.50738	0.14356	11	13.5
2019-02-08-16:02:28.800	15	39.5	2100	OFF	0	3.904	1.14848	2.65586	12.5	81
2019-02-08-16:02:29.200	16	58.5	2200	OFF	0	21.96	0.78958	4.01968	16	229.5
2019-02-08-16:02:29.600	19	0	2000	OFF	0	28.304	1.86628	0.93314	0.5	193.5
2019-02-08-16:02:30.000	16	0	1100	ON	6	-4.88	-5.52706	-3.01476	0	-58.5
2019-02-08-16:02:30.200	11	0	900	ON	10.56	-61	-8.9725	-8.9725	0	-153
2019-02-08-16:02:30.400	0	0	600	ON	7.728	-61	-1.65094	5.7424	0	-229.5

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A Different Scenario….



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System Status at Retreival

System Status at Retrieval

ECU Part Number	89170-12E20
EDR Generation	17EDR
Complete File Recorded	Yes
Ignition Cycle, Download (times)	405
Multi-Event, Number of Events (times)	N/A - 2 dependent events follow 1 independent
Time from Event 1 to 2 (sec)	N/A
Diagnostic Trouble Codes Exist	No
Location of Side Satellite Sensor 1, L	front door
Location of Side Satellite Sensor 2, L	B-Pillar
Location of Side Satellite Sensor 3, L	Not Equipped
Location of Side Satellite Sensor 4, L	C-Pillar
Location of Side Satellite Sensor 1, R	front door
Location of Side Satellite Sensor 2, R	B-Pillar
Location of Side Satellite Sensor 3, R	Not Equipped
Location of Side Satellite Sensor 4, R	C-Pillar
Location of Floor Sensor	Airbag ECU

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash Recording Status	Diagnostic Data Recording Status	Occupant Data Recording Status	Crash Info Recording Status	Time Series Recording Status
Most Recent Event	6	Frontal/Rear/Side Crash	0	Complete	Complete	Complete	Complete	Complete
1st Prior Event	5	Side Crash	29	N/A	Complete	N/A	N/A	Complete
2nd Prior Event	4	Side Crash	-284.5	N/A	Complete	N/A	N/A	Complete
3rd Prior Event	3	Side Crash	-528.5	N/A	Complete	N/A	N/A	Complete
4th Prior Event	2	Rollover	-536	Complete	Complete	Complete	Complete	Complete
5th Prior Event	1	Frontal/Rear/Side Crash	-609.5	Complete	Complete	Complete	Complete	Complete

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VCH / FFD Data

From VCH Area Files

Trigger ID	18_11	
Trigger time information	350	ms
EDR-related information	16	ms
EDR-related information	3	
Number of Overwrite of the Recording Area	1	
Trip count	413	trip
Clock Type	0	
Time count	881.6	sec
Odometer signal	4853	
Odometer unit signal	1	
HV recognition flag_header	0	
Diesel flag_header	0	

From FFD Images Set Files Area

Trip count	413	trip
Number of Overwrite	2	
Trigger Point Offset Time	400	ms
Encryption Status	1	
Pre-Trigger Image Stored Number	10	
Trigger Point Image Stored Number	1	
Post-Trigger Image Stored Number	10	
Writing completion flag	55	

From VCH Trigger Definitions

18-11	PCS operation history (when acceleration above certain amount is detected)
18-	PCS operation history (request

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Corresponding CDR Data

System Status at Event (Most Recent Event, TRG 6)	
TRG Count (times)	6
Event Type	Frontal/Rear/Side Crash
Previous Crash Type	Side Crash
Time from Previous TRG (msec)	39.0
Time from Time Zero to TRG (msec)	68.0
Event Establishment Factor	Frontal Crash
TRG Establishment Factor	Frontal Crash
Freeze Signal	OFF
Freeze Signal Factor	None
Recording Status - Front/Rear and Side Crash Info	Complete
Odometer signal (miles [km])	3,016 [4,853]
Trip count (times)	413
Time count (msec)	882,100
Time count input system	Normal

Trigger ID	18_11	
Trigger time information	350	ms
EDR-related information	16	ms
EDR-related information	3	
Number of Overwrite of the Recording Area	1	
Trip count	413	trip
Clock Type	0	
Time count	881.6	sec
Odometer signal	4853	
Odometer unit signal	1	
HV recognition flag_header	0	
Diesel flag_header	0	

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Pre-Crash – 5 To 0

Pre-Crash Data -5 to 0 Seconds (Most Recent Event, TRG 6) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-5.00	50.3 [81]	28.5	6.5	Invalid	1,300	Invalid	OFF
-4.50	50.3 [81]	28.5	7.0	Invalid	1,300	Invalid	OFF
-4.00	50.3 [81]	28.5	7.0	Invalid	1,300	Invalid	OFF
-3.50	50.3 [81]	28.5	7.0	Invalid	1,300	Invalid	OFF
-3.00	50.3 [81]	28.5	7.0	Invalid	1,300	Invalid	OFF
-2.50	50.3 [81]	28.5	7.0	Invalid	1,300	Invalid	OFF
-2.00	49.7 [80]	28.5	7.0	Invalid	1,300	Invalid	OFF
-1.50	45.4 [73]	57.5	7.5	Invalid	1,100	Invalid	OFF
-1.00	42.9 [69]	100.0	21.5	Invalid	2,000	Invalid	OFF
-0.50	29.2 [47]	0.0	2.5	Invalid	2,000	Invalid	ON
TRG(0)	13.7 [22]	0.0	0.0	Invalid	900	Invalid	ON

Pre-Crash Data -5 to 0 Seconds (Most Recent Event, TRG 6) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-5.00	OFF	OFF	0.00	-0.144	-0.98	0.0	D
-4.50	OFF	OFF	0.00	-0.359	-0.98	0.0	D
-4.00	OFF	OFF	0.00	-0.431	-0.98	0.0	D
-3.50	OFF	OFF	0.00	-0.359	-0.98	0.0	D
-3.00	OFF	OFF	0.00	-0.072	-0.98	0.0	D
-2.50	OFF	OFF	0.00	0.000	-0.98	0.0	D
-2.00	OFF	OFF	0.00	-0.287	-1.46	0.0	D
-1.50	ON	OFF	0.00	-8.973	7.32	16.5	D
-1.00	ON	OFF	0.00	-0.072	-5.86	24.0	D
-0.50	ON	OFF	6.67	-8.973	30.26	33.0	N
TRG(0)	ON	OFF	4.70	-8.973	-13.18	49.5	N

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VCH Data

Vehicle Speed [km/h]	Accelerator Opening Ratio [%]	Engine RPM Data [rpm]	Cruise Control	Shift Position Signal	Brake SW	Brake Oil Pressure [MPa]	Yaw Rate Sensor Signal [deg/s]	Longitudinal G Sensor Signal [m/s^2]	Lateral G Sensor Signal [m/s^2]	Throttle Opening Ratio [%]	Steering Signal [deg]
81	28.5	1300	0	1	OFF	0	-0.976	-0.14356	0.28712	7	0
81	28.5	1300	0	1	OFF	0	-0.976	0	0.28712	6.5	0
81	28.5	1300	0	1	OFF	0	-0.976	-0.07178	0.43068	7	0
81	28.5	1300	0	1	OFF	0	-0.488	-0.21534	0.43068	7	0
81	28.5	1300	0	1	OFF	0	-0.976	-0.3589	0.50246	7	0
80	28.5	1300	0	1	OFF	0	-0.976	-0.3589	0.43068	7	0
80	28.5	1300	0	1	OFF	0	-0.488	-0.14356	0.3589	7	0
80	28.5	1300	0	1	OFF	0	-0.976	-0.07178	0.3589	7	0
60	100	1100	0	1	OFF	0	8.784	-5.0246	-1.22026	99	22.5
75	0	3100	0	1	OFF	0	-3.904	-0.57424	0	0.5	54
71	0	2900	0	1	ON	7.008	2.928	-8.9725	-1.0767	0	54
27	0	1400	0	0	ON	7.584	-17.08	-7.68046	-1.36382	0	40.5
9	0	500	0	0	ON	5.28	-16.592	-7.24978	1.86628	1	63
1	0	100	0	0	ON	11.952	-1.952	-1.4356	0.50246	0	40.5
0	0	0	0	0	ON	8.64	-0.976	-0.3589	0.21534	0	-4.5
0	0	0	0	0	ON	9.072	-0.976	-0.3589	0.21534	0	-4.5
0	0	0	0	0	ON	9.264	-0.976	-0.3589	0.21534	0	0
0	0	0	0	0	ON	9.264	-0.976	-0.3589	0.21534	0	0
0	0	0	0	0	ON	9.216	-0.976	-0.3589	0.21534	0	0
0	0	0	0	0	ON	9.264	-0.976	-0.3589	0.21534	0	0

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